

**2003**

**HIGHER SCHOOL CERTIFICATE  
EXAMINATION**

# Physics

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 21 and 25

**Total marks – 100**

**Section I** Pages 2–28

**75 marks**

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

**Section II** Pages 29–42

**25 marks**

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

## Section I

75 marks

### Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**      $2 + 4 =$      (A) 2     (B) 6     (C) 8     (D) 9  
                                         A ☐     B ☒     C ☐     D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒     B ☒     C ☐     D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒     B ☒     C ☐     D ☐  
                                         *correct*  
                                         ↙

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- 1 The weight of an astronaut on the Moon is  $\frac{1}{6}$  of her weight on Earth.

What is the acceleration due to gravity on the Moon?

- (A)  $\left(\frac{6}{9.8}\right) \text{ms}^{-2}$   
 (B)  $\left(\frac{9.8}{6}\right) \text{ms}^{-2}$   
 (C)  $9.8 \text{ms}^{-2}$   
 (D)  $(9.8 \times 6) \text{ms}^{-2}$

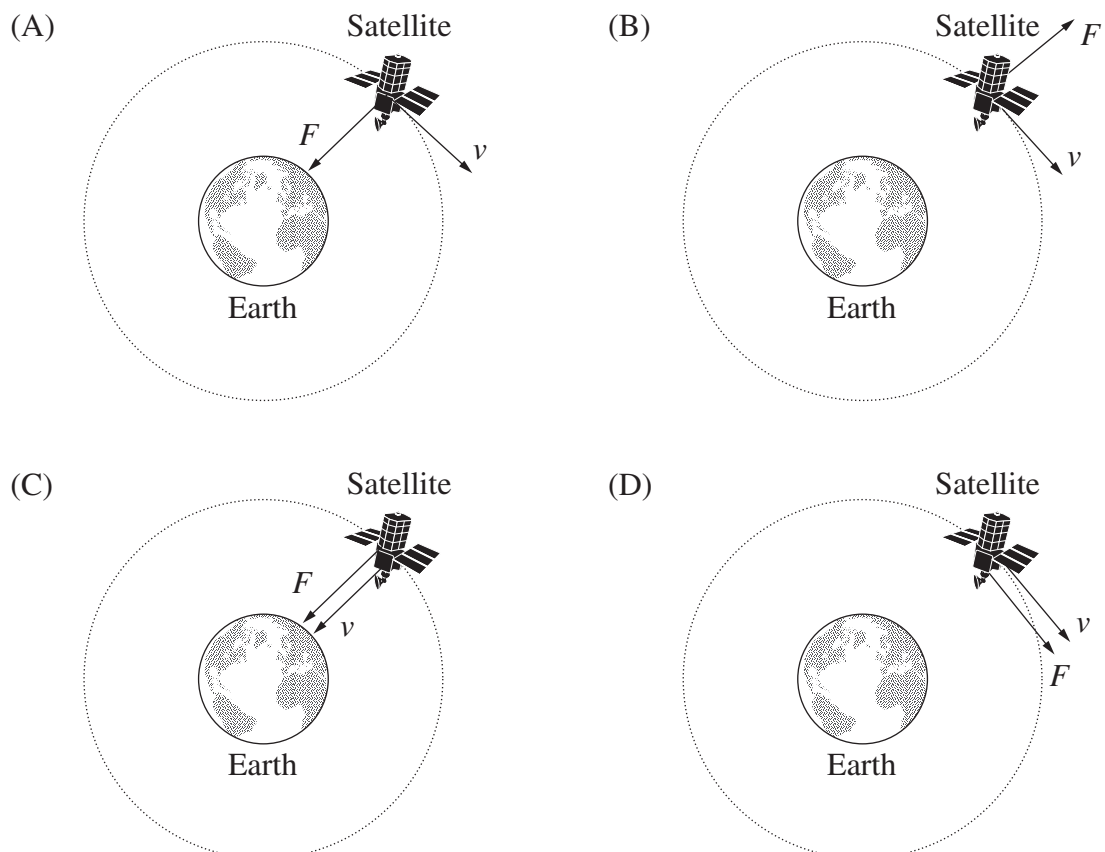
- 2 A satellite moves in uniform circular motion around Earth.

The following table shows the symbols used in the diagrams below.  
 These diagrams are NOT drawn to scale.

Key

|     |                        |
|-----|------------------------|
| $F$ | net force on satellite |
| $v$ | velocity of satellite  |

Which diagram shows the direction of  $F$  and  $v$  at the position indicated?



- 3 For a satellite moving in uniform circular motion around Earth, the centripetal force is provided by the gravitational force.

The mass of Earth is  $M_E$ .

The mass of the satellite is  $M_S$ .

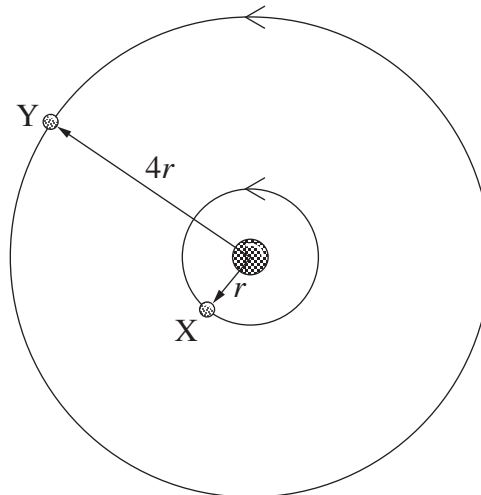
The distance of the satellite from the centre of Earth is  $d$ .

Which of the following equations should be used to calculate the speed of this satellite?

- (A)  $v = \frac{GM_E}{d}$
- (B)  $v = \sqrt{\frac{GM_E}{d}}$
- (C)  $v = \sqrt{\frac{GM_E}{d^2}}$
- (D)  $v = \sqrt{\frac{GM_E M_S}{d}}$

- 4 Two planets, X and Y, travel around a star in the same direction, in circular orbits.

Planet X completes one revolution about the star in time  $T$ . The radii of the orbits are in the ratio 1 : 4.



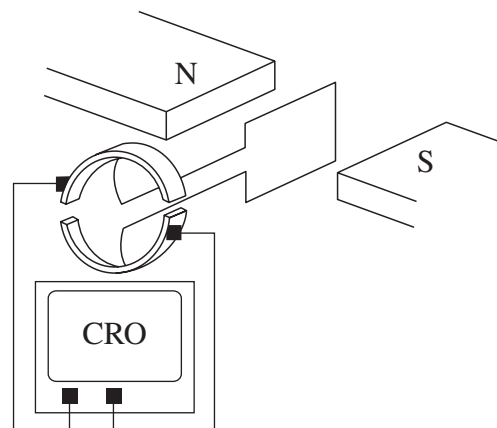
How many revolutions does planet Y make about the star in the same time  $T$ ?

- (A)  $\frac{1}{8}$  revolution
- (B)  $\frac{1}{2}$  revolution
- (C) 2 revolutions
- (D) 8 revolutions

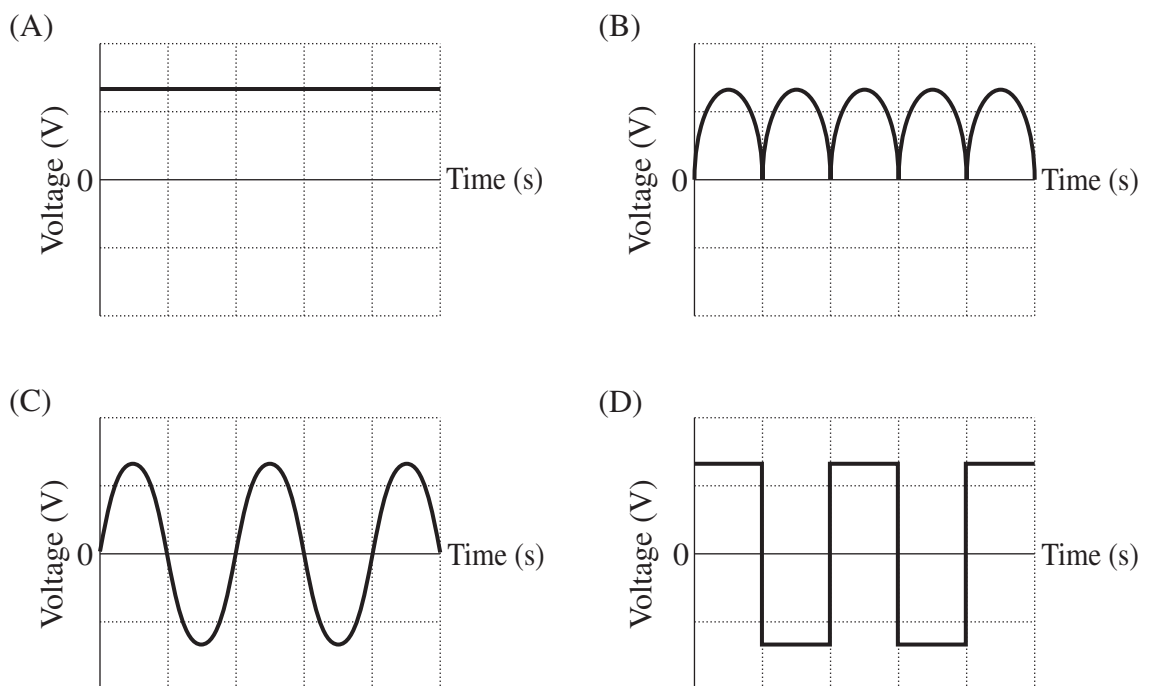
- 5 An astronaut set out in a spaceship from Earth orbit to travel to a distant star in our galaxy. The spaceship travelled at a speed of  $0.8c$ . When the spaceship reached the star the on-board clock showed the astronaut that the journey took 10 years.

An identical clock remained on Earth. What time in years had elapsed on this clock when seen from the astronaut's spaceship?

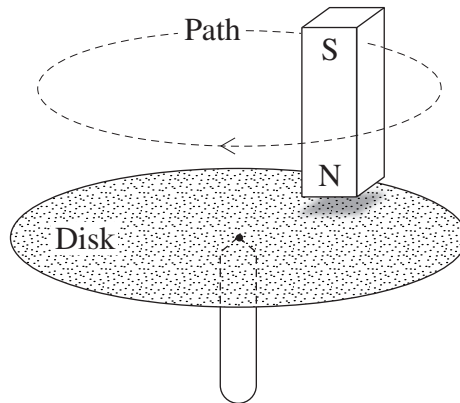
- (A) 3.6  
 (B) 6.0  
 (C) 10.0  
 (D) 16.7
- 6 The diagram shows a DC generator connected to a cathode ray oscilloscope (CRO).



What output voltage would be observed for this generator on the CRO?



- 7 A non-magnetic metal disk is balanced on a support as shown in the diagram below. The disk is initially stationary. A magnet is moved in a circular path just above the surface of the disk, without touching it.



As a result of this movement the disk begins to rotate in the same direction as the magnet.

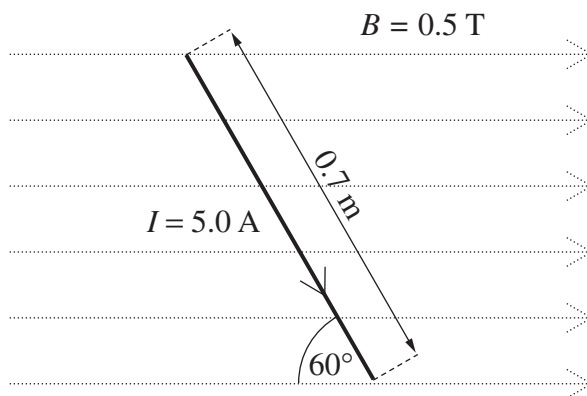
The observed effect demonstrates the principle *most* applicable to the operation of the

- (A) DC motor.
  - (B) galvanometer.
  - (C) generator.
  - (D) induction motor.
- 8 A neon sign requires a 6000 V supply for its operation. A transformer allows the neon sign to operate from a 240 V supply.

What is the ratio of the number of secondary turns to the number of primary turns for the transformer?

- (A) 1 : 40
- (B) 1 : 25
- (C) 25 : 1
- (D) 40 : 1

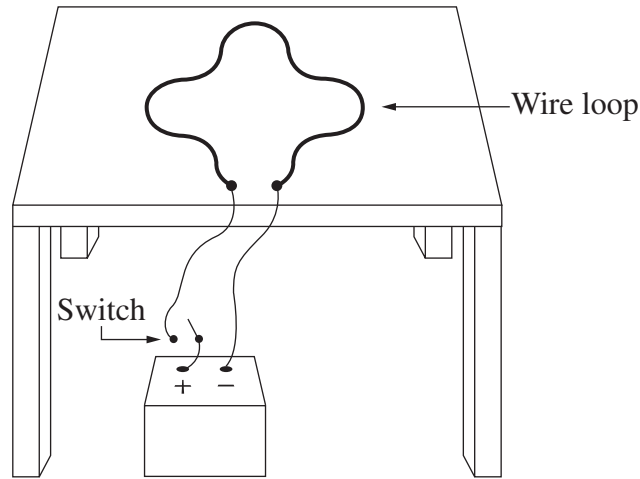
- 9 A current of 5.0 A flows in a wire that is placed in a magnetic field of 0.5 T. The wire is 0.7 m long and is at an angle of  $60^\circ$  to the field.



What is the approximate magnitude of the force on the wire?

- (A) 0 N
- (B) 0.9 N
- (C) 1.5 N
- (D) 1.8 N

- 10 A flexible wire loop is lying on a frictionless table made from an insulating material. The wire can slide around horizontally on the table and change shape freely, but it cannot move vertically. The loop is connected to a power supply, a switch and two terminals fixed to the table as shown.



When the switch is closed, a current  $I$  flows around the loop.

Which of the following diagrams most closely represents the final shape of the loop after the switch is closed?

- (A) (B) (C) (D)
- Diagram (A) shows a circular loop with two terminals at the bottom. An arrow labeled  $I$  indicates a clockwise current flow. Diagram (B) shows a cross-shaped loop with two terminals at the bottom. An arrow labeled  $I$  indicates an upward current flow. Diagram (C) shows a four-lobed loop, identical in shape to the initial state, with two terminals at the bottom. An arrow labeled  $I$  indicates a clockwise current flow. Diagram (D) shows a vertically elongated, narrow loop with two terminals at the bottom. An arrow labeled  $I$  indicates an upward current flow.

- 11 Which of the following did the Braggs investigate using X-ray diffraction?
- (A) Cathode rays
  - (B) Crystal structure
  - (C) Photoelectric effect
  - (D) Superconductivity
- 12 In a first-hand investigation that you performed, you used a discharge tube containing a Maltese Cross. You would have observed an image similar to the one shown below.



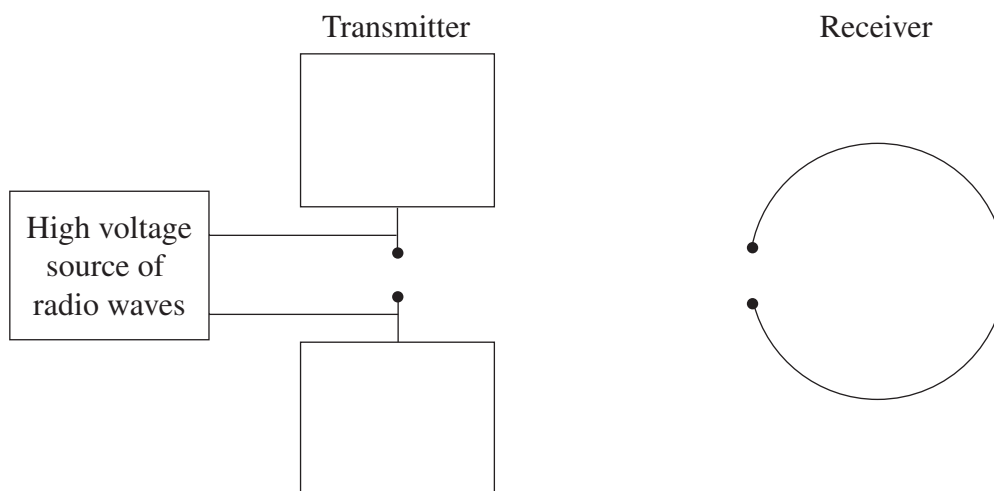
Which of the following statements is a valid conclusion from the observations made in this Maltese Cross investigation?

- (A) Cathode rays pass through glass.
- (B) Cathode rays pass through metals.
- (C) Cathode rays are charged particles.
- (D) Cathode rays travel in straight lines.

- 13** An  $n$ -type semiconductor is produced when silicon crystal is doped with small quantities of phosphorus.

How will this doping change the crystal's electrical conductivity?

- (A) The conductivity will decrease because there are fewer holes in the valence band.
  - (B) The conductivity will increase because there are more holes in the valence band.
  - (C) The conductivity will decrease because there are fewer electrons in the conduction band.
  - (D) The conductivity will increase because there are more electrons in the conduction band.
- 14** Heinrich Hertz used a set-up similar to the one shown below to investigate the production and detection of electromagnetic radiation.

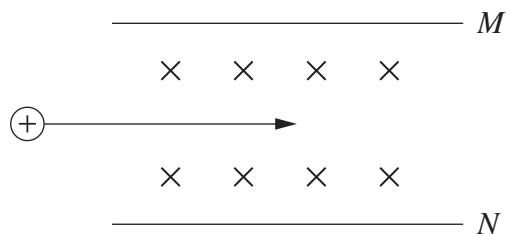


A glass sheet was placed between the transmitter and receiver.

Which of the following observations is consistent with the photoelectric effect that Hertz produced?

- (A) Radio waves were blocked when the glass sheet was in place.
- (B) Ultraviolet waves were blocked when the glass sheet was in place.
- (C) The maximum spark length was longer when the glass sheet was in place.
- (D) The maximum spark length was shorter when the glass sheet was in place.

- 15 A positively-charged ion travelling at  $250 \text{ m s}^{-1}$  is fired between two parallel charged plates,  $M$  and  $N$ . There is also a magnetic field present in the region between the two plates. The direction of the magnetic field is into the page as shown. The ion is travelling perpendicular to both the electric and the magnetic fields.



The electric field between the plates has a magnitude of  $200 \text{ V m}^{-1}$ . The magnetic field is adjusted so that the ion passes through undeflected.

What is the magnitude of the adjusted magnetic field, and the polarity of the  $M$  terminal relative to the  $N$  terminal?

|     | <i>Magnitude of magnetic field (teslas)</i> | <i>Polarity of <math>M</math> relative to <math>N</math></i> |
|-----|---------------------------------------------|--------------------------------------------------------------|
| (A) | 0.8                                         | positive                                                     |
| (B) | 0.8                                         | negative                                                     |
| (C) | 1.25                                        | positive                                                     |
| (D) | 1.25                                        | negative                                                     |

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Centre Number

**Section I (continued)**

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**Part B – 60 marks**

Student Number

**Attempt Questions 16–27**

**Allow about 1 hour and 45 minutes for this part**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

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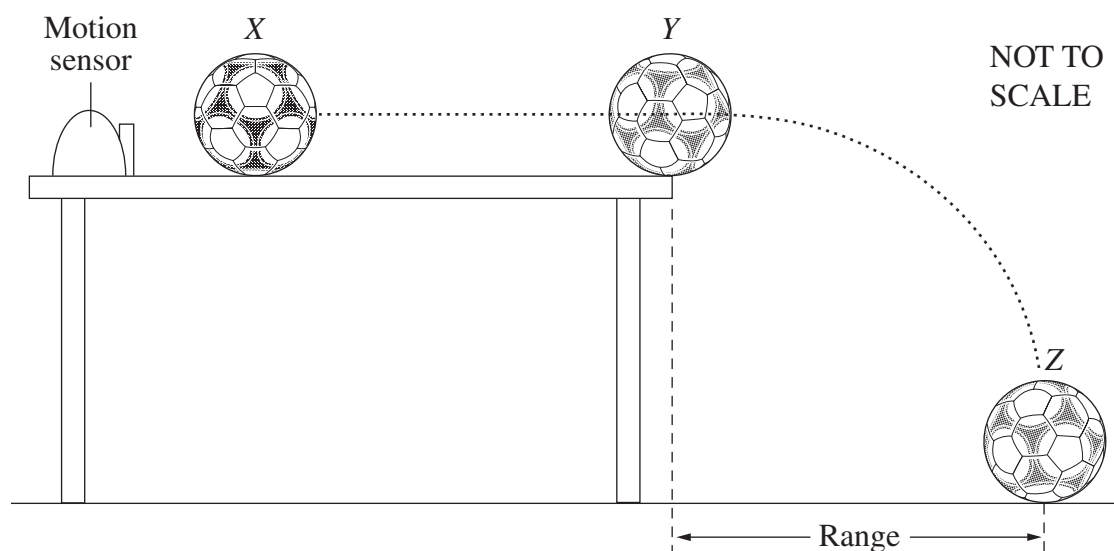
**Question 16 (6 marks)**

**Please turn over**

**Question 16** (6 marks)

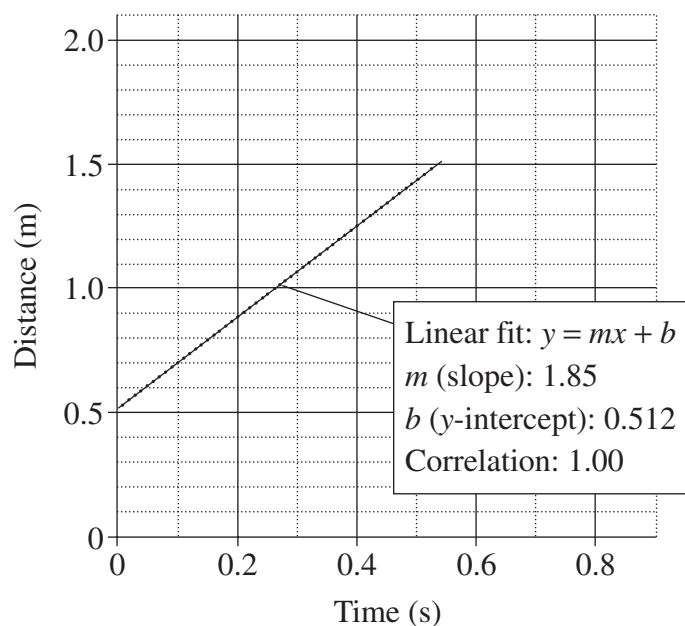
A student performed a first-hand investigation to examine projectile motion.

A ball resting on a horizontal table was given an initial push at X, resulting in the ball following the path XYZ as shown.



A data logger used the motion sensor to measure the horizontal distance to the ball. When the ball was at position Y, a distance of 1.50 m from the motion sensor, it left the edge of the table.

In the first trial, the range was 0.60 m. The graph below was obtained from the data logger.



**Question 16 continues on page 15**

Question 16 (continued)

- (a) For this trial, determine the horizontal speed of the ball as it left the edge of the table. 1

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- (b) The experiment was repeated with the ball leaving the table at different speeds. Graph the relationship between the range and the horizontal speed at  $Y$ . Identify on your graph the results from the first trial. 3



- (c) The apparatus described in this first-hand investigation was used to carry out an identical experiment on another planet where the acceleration due to gravity is less than that on Earth. 2

The horizontal speed of the ball as it left the table on the planet was the same as in part (a). Compare the range of the ball on the planet to that on Earth. Explain your answer.

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**End of Question 16**

**Question 17** (6 marks)

A satellite of mass 150 kg is launched from Earth's surface into a uniform circular orbit of radius  $7.5 \times 10^6$  m.

- (a) Calculate the magnitude of the gravitational potential energy  $E_p$  of the satellite. **1**

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- (b) From this uniform circular orbit, the satellite can escape Earth's gravitational field when its kinetic energy is equal to the magnitude of the gravitational potential energy. **3**

Use this relationship to calculate the escape velocity of the satellite.

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- (c) Discuss the effect of Earth's rotational motion on the launch of this satellite. **2**

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## Section I – Part B (continued)

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Student Number

Marks

## Question 18 (6 marks)

Michelson and Morley set up an experiment to measure the velocity of Earth relative to the aether.

- (a) Outline TWO features of the aether model for the transmission of light. 2

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- (b) Recount the Michelson and Morley experiment, which attempted to measure the relative velocity of Earth through the aether, and describe the results they anticipated. 4

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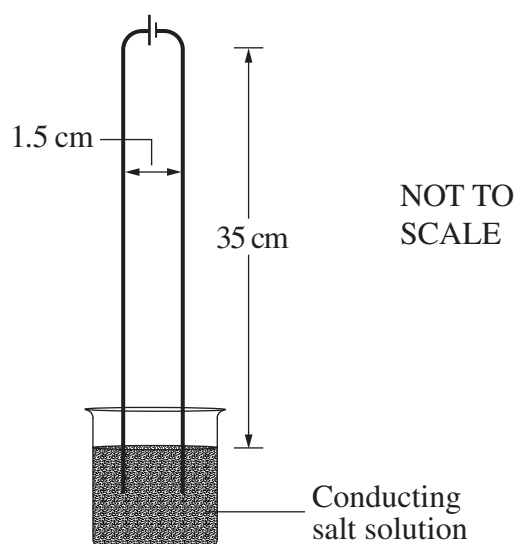
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**Question 19** (3 marks)

Two straight copper wires are suspended so that their lower ends dip into a conducting salt solution in a beaker as shown. The length of the straight section of each wire above the conducting salt solution is 35 cm and they are placed 1.5 cm apart. The ends of the wire do not touch the bottom of the beaker. The two wires are connected to a DC power supply.

**3**

A current of 2 amperes flows from the battery. Calculate the magnitude and direction of the initial force on each wire.

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**Question 20** (4 marks)

Two solenoids (coils) with hollow cores are suspended using string so that they are hanging in the positions shown below. The solenoids are free to move in a pendulum motion.

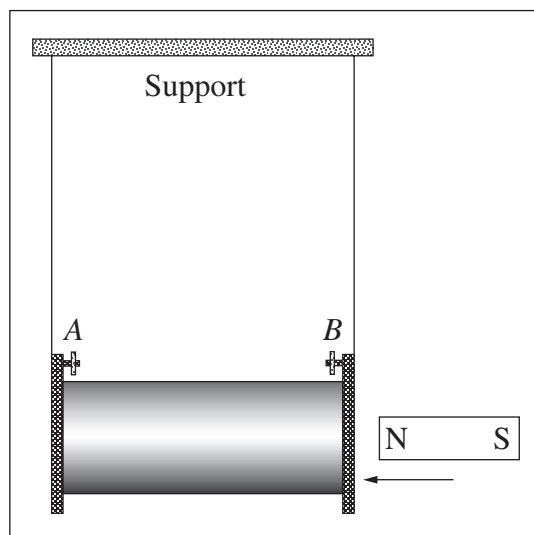
**4**

Figure 1 – First investigation

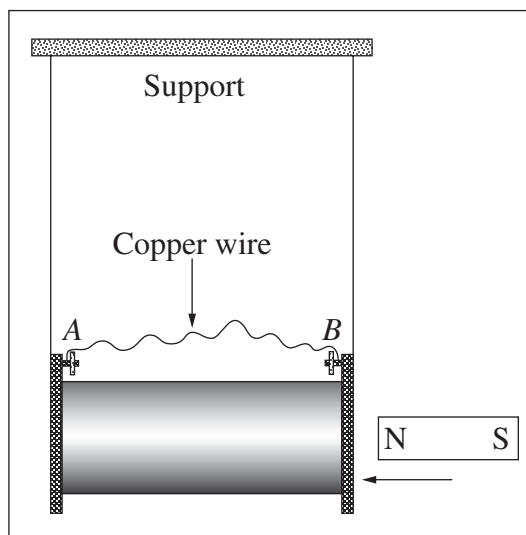


Figure 2 – Second investigation

In the first investigation shown in Figure 1, a strong bar magnet is moved towards the solenoid until the north end of the magnet enters the solenoid and then the motion of the magnet is stopped.

In the second investigation, shown in Figure 2, a thick copper wire is connected between the two terminals, A and B, at the ends of the solenoid. The motion of the magnet is repeated exactly in this second investigation.

Explain the effect of the motion of the magnet on the solenoid in the two investigations.

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Centre Number

## Section I – Part B (continued)

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Student Number

Marks

## Question 21 (5 marks)

- (a) Explain the relationship between the current in the primary coil and the current in the secondary coil of an ideal step-down transformer in relation to the conservation of energy. 3

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- (b) Explain why a transformer will work in an AC circuit but not in a DC circuit. 2

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**Question 22** (5 marks)

Describe a first-hand investigation to demonstrate the effect on a generated electric current when the strength of the magnet is varied.

**5**

In your description, include:

- a labelled sketch of the experimental set-up;
- how you varied the magnetic field strength;
- how other variables were controlled.

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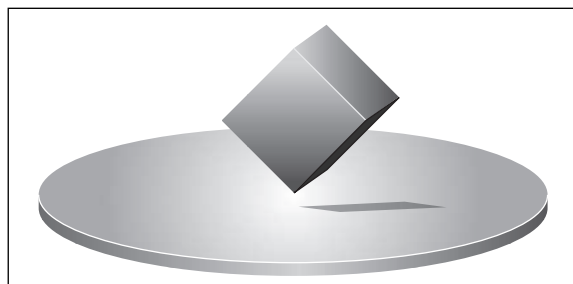
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**Question 23** (6 marks)

- (a) The following image shows a magnet hovering above a superconducting disk. **3**



Explain why the magnet is able to hover above the superconductor.

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- (b) Compare the model for the conduction of electricity in metals at room temperature with the model for conduction of electricity in superconductors below the critical temperature. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

**Marks**

**Question 24** (4 marks)

Outline Thomson's experiment to measure the charge/mass ratio of an electron.

**4**

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**Question 25** (5 marks)

A physics student was conducting an investigation on the photoelectric effect. The student used an infrared laser with a wavelength of  $1.55 \times 10^{-6} \text{ m}$  for this investigation.

- (a) Calculate the energy of a photon from this laser. **2**

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- (b) When the laser light was shone onto a photo-cell, no current was detected. The student increased the intensity of the light but still detected no current. **3**

Explain this observation.

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**Question 26** (6 marks)

Describe Einstein's contributions to Special Relativity and to Quantum Theory and how these contributions changed the direction of scientific thinking in the Twentieth Century.

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**Question 27** (4 marks)

In a particle accelerator called a synchrotron, magnetic fields are used to control the motion of an electron so that it follows a circular path of fixed radius.

**4**

Describe the changes required in the magnetic field to accelerate an electron to near the speed of light. Support your answer with appropriate mathematical relationships.

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## Physics

### Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

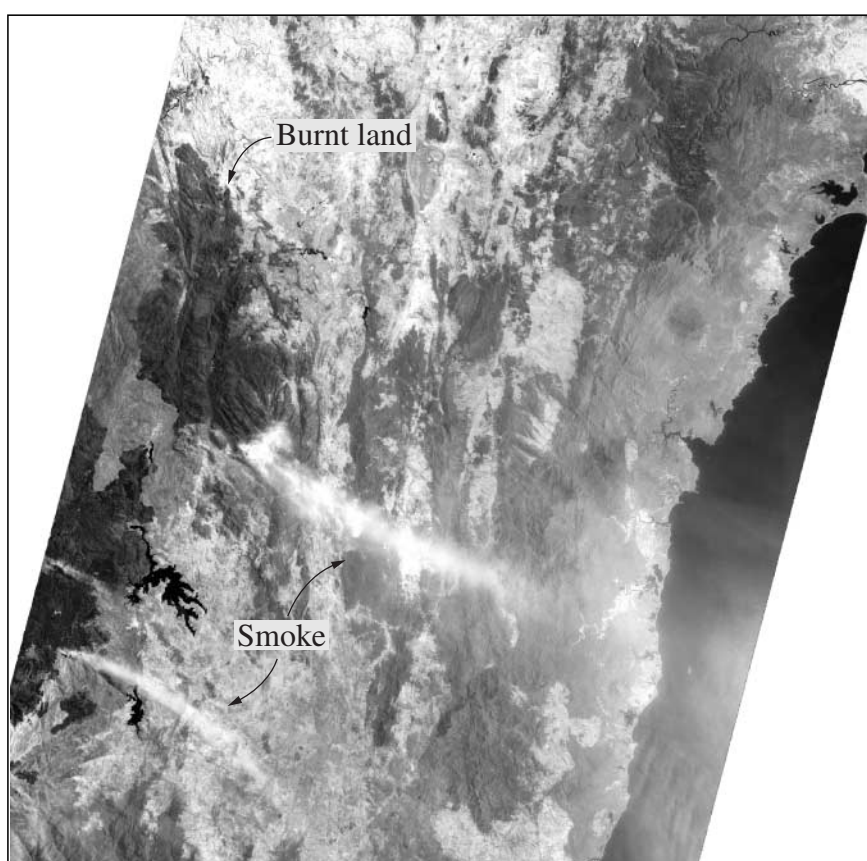
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|                                            | Pages |
|--------------------------------------------|-------|
| Question 28    Geophysics .....            | 31–33 |
| Question 29    Medical Physics .....       | 34–35 |
| Question 30    Astrophysics .....          | 36–38 |
| Question 31    From Quanta to Quarks ..... | 39–40 |
| Question 32    The Age of Silicon .....    | 41–42 |

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**Question 28 — Geophysics (25 marks)**

- |     |      |                                                                                                                                                                                            |          |
|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (a) | (i)  | Identify THREE principal methods used by geophysicists to investigate the structure of Earth and the properties of Earth materials.                                                        | <b>1</b> |
|     | (ii) | Describe the role that geophysicists play in the monitoring of nuclear test-ban treaties.                                                                                                  | <b>2</b> |
| (b) |      | Summarise the geophysical evidence that supports the theory of plate tectonics.                                                                                                            | <b>3</b> |
| (c) | (i)  | Describe how absorption and reflection of radiation can provide information about a reflecting surface.                                                                                    | <b>2</b> |
|     | (ii) | The picture below shows a satellite image of a bushfire burning in a forested area. Images such as the one below can be used as a part of the process of monitoring changes in vegetation. | <b>3</b> |



Explain how remote-sensing techniques can be used to monitor the spread of a bushfire, and the regrowth of vegetation in regions affected by a bushfire.

**Question 28 continues on page 32**

## Question 28 (continued)

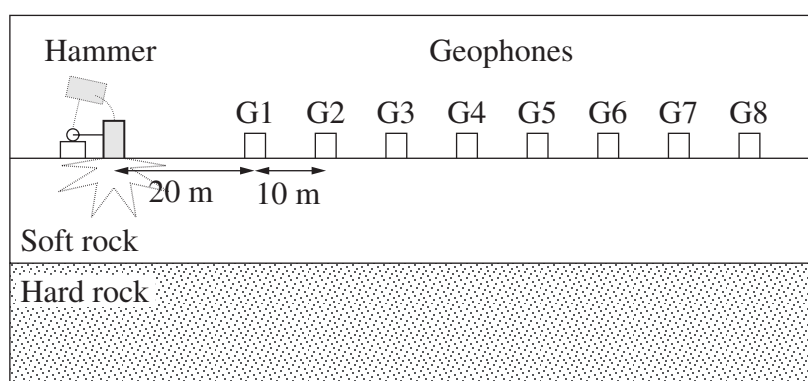
- (d) (i) Outline the structure and function of a geophone.

2

- (ii) The method of seismic refraction is depicted in the diagram below. A series of eight geophones, G1 to G8, are arranged in a straight line along level ground. They are each separated by a distance of 10 m.

At a distance of 20 m from the first geophone, a hammer is used to strike the ground to produce seismic waves.

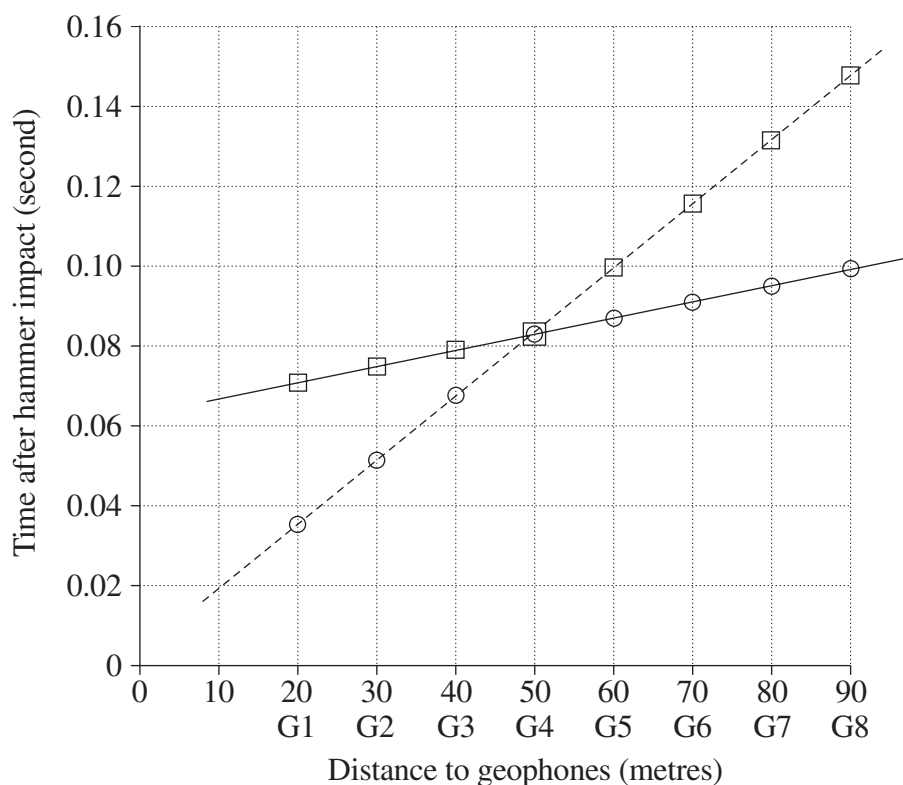
The geophones are attached to a seismograph that records the time of arrival of the waves after the hammer strikes the ground.



The data from the geophones are analysed and the arrival times of the direct and refracted waves that reach each geophone are recorded. These data are shown in the graph on page 33. On the graph, a circle represents the arrival of the first wave to reach a geophone, and a square represents the arrival time of the second wave to reach a geophone. The points on the graph associated with the direct seismic wave and the refracted seismic wave are shown.

**Question 28 continues on page 33**

## Question 28 (continued)



## Legend

- Time of arrival of first wave at geophone
- Time of arrival of second wave at geophone
- Refracted wave
- Direct wave

- (1) Explain why the line for the refracted wave crosses the line for the direct wave on the graph. 2
  - (2) From the graph, calculate the speed of the direct wave in the soft rock layer. 2
- (e) Outline the application of Newton's theory of universal gravitation to the field of geophysics, and discuss how information obtained from gravity surveys has led to a greater understanding of the structure of Earth. 8

**End of Question 28**

**Question 29 — Medical Physics (25 marks)**

- |     |                                                                                                                       |          |
|-----|-----------------------------------------------------------------------------------------------------------------------|----------|
| (a) | (i) Identify the property of the hydrogen nucleus that makes it useful in magnetic resonance imaging.                 | <b>1</b> |
|     | (ii) Describe how X-rays are produced when electrons strike the anode in an X-ray tube.                               | <b>2</b> |
| (b) | Outline the production of gamma rays and their use in the diagnostic procedure of positron emission tomography (PET). | <b>3</b> |
| (c) | This question refers to the bone scan of a person with cancer, and a chest X-ray of a healthy person.                 |          |



- |      |                                                                                       |          |
|------|---------------------------------------------------------------------------------------|----------|
| (i)  | Compare how radiation is used to produce a bone scan image and an X-ray image.        | <b>3</b> |
| (ii) | Describe how a bone scan is able to provide information that an X-ray cannot provide. | <b>2</b> |

**Question 29 continues on page 35**

## Question 29 (continued)

- (d) The table below shows the speed of sound in, and density of, several different tissues.

| <i>Tissue</i> | <i>Speed of sound in tissue (m s<sup>-1</sup>)</i> | <i>Density (kg m<sup>-3</sup>)</i> |
|---------------|----------------------------------------------------|------------------------------------|
| Fat           | 1450                                               | 952                                |
| Blood         | 1570                                               | 1025                               |
| Kidney        | 1560                                               | 1038                               |
| Liver         | 1550                                               | 1065                               |
| Muscle        | 1580                                               | 1076                               |

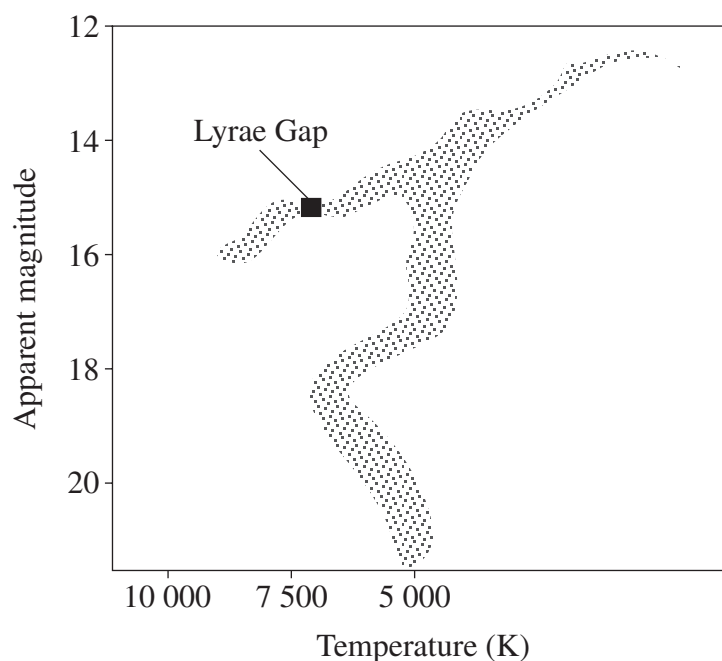
- (i) Calculate the acoustic impedance of kidney tissue. **1**
- (ii) Ultrasound travelling through kidney tissue in the body encounters a different type of tissue. Identify the type of tissue that will result in the greatest proportion of the incident pulse being reflected at the boundary between the kidney and the other tissue. Justify your choice. **2**
- (iii) Describe the properties of ultrasound that led to its use in the measurement of bone density. **3**
- (e) An understanding of the properties of electrons, and our ability to control their behaviour, have played key roles in the development of CAT scans and positron emission tomography imaging technologies. **8**

Justify this statement with reference to the production and display of images used for medical diagnosis.

**End of Question 29**

**Question 30 — Astrophysics (25 marks)**

- (a) (i) Define the term *resolution* of a telescope. 1
- (ii) Describe ONE method by which the resolution of a ground-based system can be improved. 2
- (b) An H-R diagram for the globular cluster M3 is shown below. 3



The stars in the Lyrae gap have an absolute magnitude of 0.6. Use this information and their position on the H-R diagram to determine the distance of M3 from Earth.

**Question 30 continues on page 37**

Question 30 (continued)

- (c) The diagram below is a comparison of the spectrum of quasar 3C 273 and a spectrum from a light source on Earth.

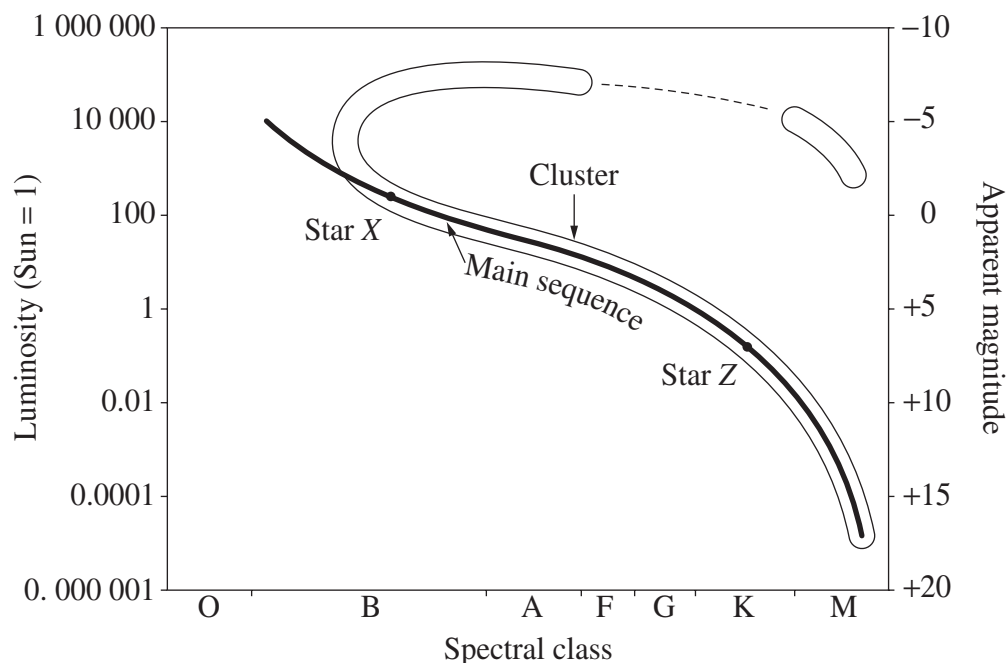


- |                                                                 |                                                                                                                                                         |          |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (i)                                                             | From this comparison, identify the feature of the quasar spectrum that is representative of the spectra produced by quasars.                            | <b>1</b> |
| (ii) The spectra above are both examples of absorption spectra. |                                                                                                                                                         |          |
| (1)                                                             | Account for the production of a star's absorption spectrum.                                                                                             | <b>2</b> |
| (2)                                                             | Describe how a spectrum from a star can provide information on the surface temperature of that star. Give a specific example to illustrate your answer. | <b>2</b> |

**Question 30 continues on page 38**

Question 30 (continued)

(d) The H-R diagram for a cluster is shown below.



- (i) Why is the cluster considered young? 1
- (ii) Stars X and Z are both part of the same cluster but have different main sequence nuclear reactions and different evolutionary pathways.
- (1) Contrast the fusion reactions in star X and star Z. 2
- (2) Predict TWO possible evolutionary pathways for star X. 3
- (e) Evaluate the impact of studying the visible spectrum of light on our understanding of celestial objects. 8

End of Question 30

**Question 31 — From Quanta to Quarks (25 marks)**

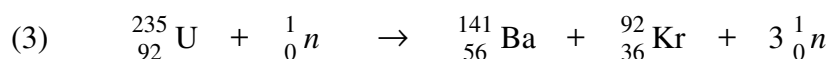
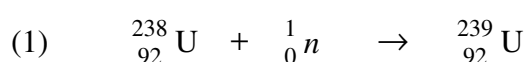
- (a) (i) Identify the structure of the Rutherford model of the atom. **1**
- (ii) Describe how Bohr refined Rutherford's model of the hydrogen atom. **2**
- (b) The table below shows the different types of quarks and their charge. **3**

| <i>Quark</i> | <i>Charge</i>   |
|--------------|-----------------|
| Up           | $+\frac{2}{3}e$ |
| Down         | $-\frac{1}{3}e$ |
| Strange      | $-\frac{1}{3}e$ |
| Charm        | $+\frac{2}{3}e$ |
| Bottom       | $-\frac{1}{3}e$ |
| Top          | $+\frac{2}{3}e$ |

The standard model of matter says that protons and neutrons are composed of up and down quarks. There are three quarks in each particle.

Compare protons and neutrons in terms of their quark composition.

- (c) The equations shown below describe three different types of transmutation reactions involving uranium.

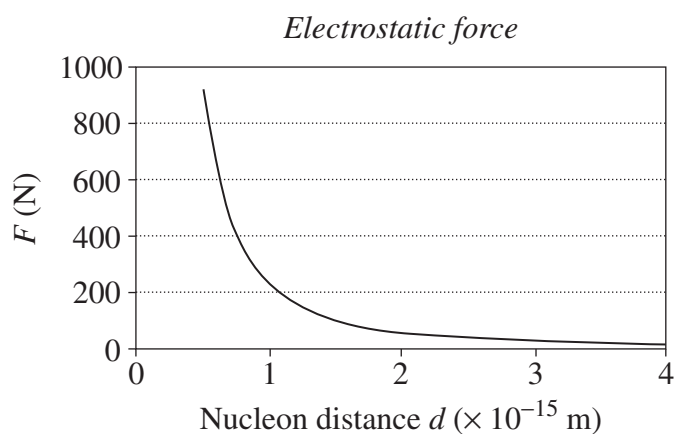
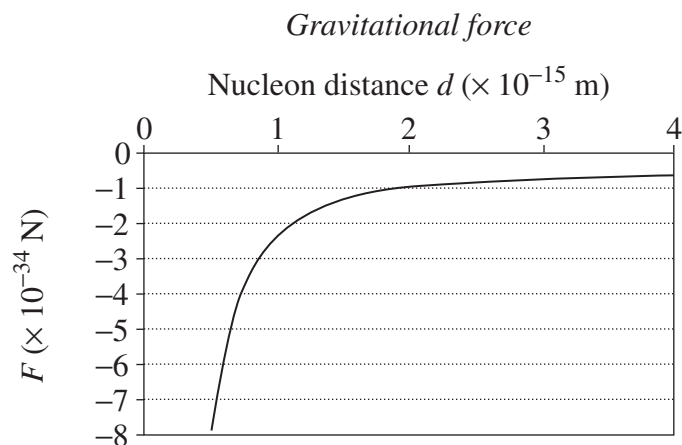


- (i) Identify which reaction is naturally occurring, and justify your answer. **2**
- (ii) Identify ONE transmutation reaction above that has a practical application, and describe the application. **3**

**Question 31 continues on page 40**

Question 31 (continued)

- (d) The two graphs below show the gravitational and electrostatic forces acting between two protons in the nucleus of an atom.

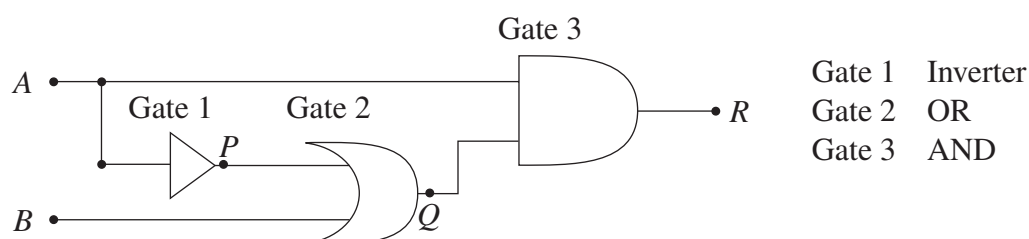


- (i) If the distance between protons in a nucleus is  $1.0 \times 10^{-15}$  m, determine both the gravitational and the electrostatic force at this distance. 2
- (ii) Explain why these two forces cannot explain the stability of the nucleus, and why there is a need for the strong nuclear force. 2
- (iii) Describe TWO properties of the strong nuclear force. 2
- (e) Describe the requirements for a nuclear fission explosion, and describe how these are controlled in a nuclear reactor. 8

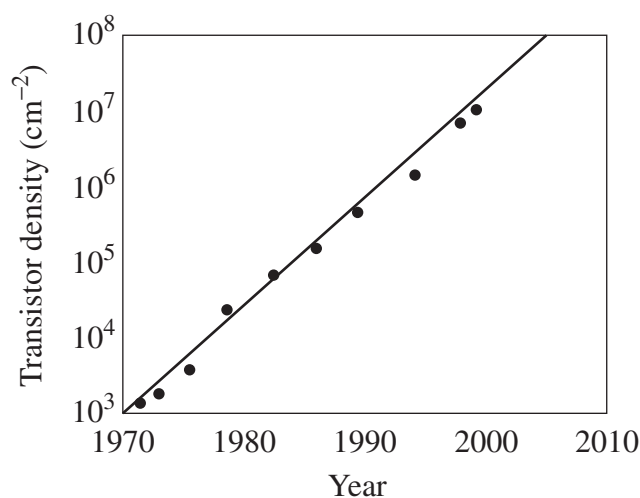
**End of Question 31**

**Question 32 — The Age of Silicon (25 marks)**

- (a) (i) Identify ONE electronic system that is digital, and ONE electronic system that is analogue. **1**
- (ii) Use diagrams to describe the variation between digital and analogue voltage outputs with time. **2**
- (b) Construct a truth table showing the outputs at  $P$ ,  $Q$  and  $R$  for each of the possible input states of  $A$  and  $B$  in the following circuit. **3**



- (c) The graph below shows how the density of transistors on a silicon chip has increased over the last 30 years.

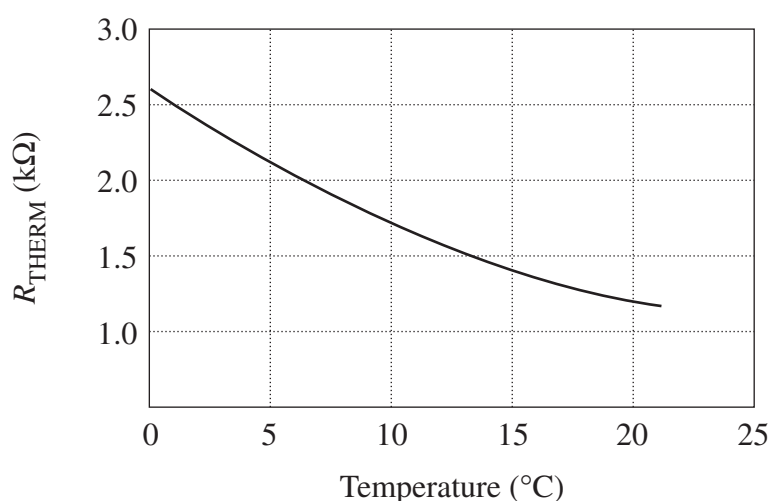
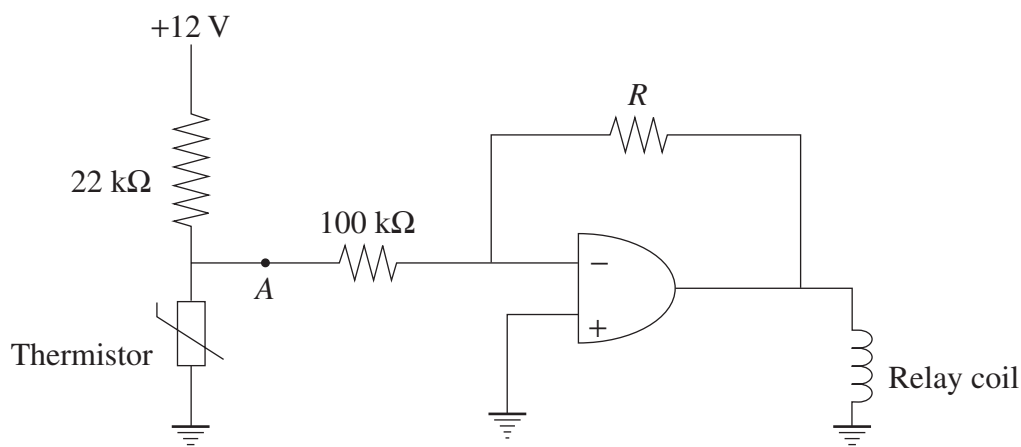


- (i) Use the data in the graph to predict the change in computer performance from 1970 to 2005. Justify your answer. **3**
- (ii) Discuss the validity of using the graph to predict computer performance up to 2060. **2**

**Question 32 continues on page 42**

Question 32 (continued)

- (d) The circuit below uses a thermistor as a temperature sensor to control the operation of a relay. The relay will close when the voltage across the relay coil is greater than 6 volts. The resistance of the thermistor,  $R_{\text{THERM}}$ , is given in the graph.



- (i) Calculate the voltage at point A at a temperature of 15°C. Neglect the effect of the 100 kΩ resistor and the operational amplifier on the voltage at point A. 3
- (ii) Determine the value of  $R$  so that the relay will close only when the temperature falls below 15°C. 3
- (e) Describe and compare the physical principles underlying the operation of input and output transducers. Use an analogue ammeter and a solar cell as examples. 8

End of paper

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## DATA SHEET

|                                                                     |                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------|
| Charge on electron, $q_e$                                           | $-1.602 \times 10^{-19} \text{ C}$                            |
| Mass of electron, $m_e$                                             | $9.109 \times 10^{-31} \text{ kg}$                            |
| Mass of neutron, $m_n$                                              | $1.675 \times 10^{-27} \text{ kg}$                            |
| Mass of proton, $m_p$                                               | $1.673 \times 10^{-27} \text{ kg}$                            |
| Speed of sound in air                                               | $340 \text{ m s}^{-1}$                                        |
| Earth's gravitational acceleration, $g$                             | $9.8 \text{ m s}^{-2}$                                        |
| Speed of light, $c$                                                 | $3.00 \times 10^8 \text{ m s}^{-1}$                           |
| Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$ | $2.0 \times 10^{-7} \text{ N A}^{-2}$                         |
| Universal gravitational constant, $G$                               | $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$          |
| Mass of Earth                                                       | $6.0 \times 10^{24} \text{ kg}$                               |
| Planck constant, $h$                                                | $6.626 \times 10^{-34} \text{ J s}$                           |
| Rydberg constant, $R$ (hydrogen)                                    | $1.097 \times 10^7 \text{ m}^{-1}$                            |
| Atomic mass unit, $u$                                               | $1.661 \times 10^{-27} \text{ kg}$<br>$931.5 \text{ MeV}/c^2$ |
| 1 eV                                                                | $1.602 \times 10^{-19} \text{ J}$                             |
| Density of water, $\rho$                                            | $1.00 \times 10^3 \text{ kg m}^{-3}$                          |
| Specific heat capacity of water                                     | $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$           |

## FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v-u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

## FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left( \frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

| PERIODIC TABLE OF THE ELEMENTS |                                |                                |                                |                                |                                 |                                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|---------------------------------|-----------------------------------|--------------------------------|------------------------------|--------------------------------|-------------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|--------------------------------|---------------------------------|---------------------------------|------------------------------|------------|--|--|--|
| KEY                            |                                |                                |                                |                                |                                 |                                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
|                                |                                | Atomic Number                  |                                | 79<br>Au<br>197.0              |                                 | Symbol of element                 |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
|                                |                                | Atomic Weight                  |                                | Gold                           |                                 | Name of element                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
| 1<br>H<br>1.008<br>Hydrogen    | 4<br>Be<br>9.012<br>Beryllium  | 5<br>B<br>10.81<br>Boron       | 6<br>C<br>12.01<br>Carbon      | 7<br>N<br>14.01<br>Nitrogen    | 8<br>O<br>16.00<br>Oxygen       | 9<br>F<br>19.00<br>Fluorine       | 10<br>Ne<br>20.18<br>Neon      |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
| 11<br>Na<br>22.99<br>Sodium    | 12<br>Mg<br>24.31<br>Magnesium | 13<br>Al<br>26.98<br>Aluminium | 14<br>Si<br>28.09<br>Silicon   | 15<br>P<br>30.97<br>Phosphorus | 16<br>S<br>32.07<br>Sulfur      | 17<br>Cl<br>35.45<br>Chlorine     | 18<br>Ar<br>39.95<br>Argon     |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |            |  |  |  |
| 19<br>K<br>39.10<br>Potassium  | 20<br>Ca<br>40.08<br>Calcium   | 21<br>Sc<br>44.96<br>Scandium  | 22<br>Ti<br>47.87<br>Titanium  | 23<br>V<br>50.94<br>Vanadium   | 24<br>Cr<br>52.00<br>Chromium   | 25<br>Mn<br>54.94<br>Manganese    | 26<br>Fe<br>55.85<br>Iron      | 27<br>Co<br>58.93<br>Cobalt  | 28<br>Ni<br>58.69<br>Nickel    | 29<br>Cu<br>63.55<br>Copper   | 30<br>Zn<br>65.39<br>Zinc    | 31<br>Ga<br>69.72<br>Gallium | 32<br>Ge<br>72.61<br>Germanium | 33<br>As<br>74.92<br>Arsenic  | 34<br>Se<br>78.96<br>Selenium  | 35<br>Br<br>79.90<br>Bromine    | 36<br>Kr<br>83.80<br>Krypton    |                              |            |  |  |  |
| 37<br>Rb<br>85.47<br>Rubidium  | 38<br>Sr<br>87.62<br>Strontium | 39<br>Y<br>88.91<br>Yttrium    | 40<br>Zr<br>91.22<br>Zirconium | 41<br>Nb<br>92.91<br>Niobium   | 42<br>Mo<br>95.94<br>Molybdenum | 43<br>Tc<br>[98.91]<br>Technetium | 44<br>Ru<br>101.1<br>Ruthenium | 45<br>Rh<br>102.9<br>Rhodium | 46<br>Pd<br>106.4<br>Palladium | 47<br>Ag<br>107.9<br>Silver   | 48<br>Cd<br>112.4<br>Cadmium | 49<br>In<br>114.8<br>Indium  | 50<br>Sn<br>118.7<br>Tin       | 51<br>Sb<br>121.8<br>Antimony | 52<br>Te<br>127.6<br>Tellurium | 53<br>I<br>126.9<br>Iodine      | 54<br>Xe<br>131.3<br>Xenon      |                              |            |  |  |  |
| 55<br>Cs<br>132.9<br>Caesium   | 56<br>Ba<br>137.3<br>Barium    | 57–71<br>Lanthanides           |                                | 72<br>Hf<br>178.5<br>Hafnium   | 73<br>Ta<br>180.9<br>Tantalum   | 74<br>W<br>183.8<br>Tungsten      | 75<br>Re<br>186.2<br>Rhenium   | 76<br>Os<br>190.2<br>Osmium  | 77<br>Ir<br>192.2<br>Iridium   | 78<br>Pt<br>195.1<br>Platinum | 79<br>Au<br>197.0<br>Gold    | 80<br>Hg<br>200.6<br>Mercury | 81<br>Tl<br>204.4<br>Thallium  | 82<br>Pb<br>207.2<br>Lead     | 83<br>Bi<br>209.0<br>Bismuth   | 84<br>Po<br>[210.0]<br>Polonium | 85<br>At<br>[210.0]<br>Astatine | 86<br>Rn<br>[222.0]<br>Radon |            |  |  |  |
| 87<br>Fr<br>[223.0]            | 88<br>Ra<br>[226.0]            | 89–103<br>Actinides            |                                | 104<br>Rf<br>[261.1]           | 105<br>Db<br>[262.1]            | 106<br>Sg<br>[263.1]              | 107<br>Bh<br>[264.1]           | 108<br>Hs<br>[265.1]         | 109<br>Mt<br>[268]             | 110<br>Uun<br>—               | 111<br>Uuu<br>—              | 112<br>Uub<br>—              | 113<br>Uuq<br>—                | 114<br>Uuq<br>—               | 115<br>—                       | 116<br>Uuh<br>—                 | 117<br>—                        | 118<br>Uuo<br>—              | Ununoctium |  |  |  |

Lanthanides

|                                |                             |                                   |                                |                                   |                               |                               |                                 |                              |                                 |                              |                             |                              |                                |                               |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|-----------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>[146.9]<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.0<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|-----------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|

Actinides

|                                 |                              |                                   |                             |                                  |                                  |                                  |                               |                                  |                                    |                                    |                                 |                                     |                                  |                                    |
|---------------------------------|------------------------------|-----------------------------------|-----------------------------|----------------------------------|----------------------------------|----------------------------------|-------------------------------|----------------------------------|------------------------------------|------------------------------------|---------------------------------|-------------------------------------|----------------------------------|------------------------------------|
| 89<br>Ac<br>[227.0]<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>[237.0]<br>Neptunium | 94<br>Pu<br>[239.1]<br>Plutonium | 95<br>Am<br>[241.1]<br>Americium | 96<br>Cm<br>[244.1]<br>Curium | 97<br>Bk<br>[249.1]<br>Berkelium | 98<br>Cf<br>[252.1]<br>Californium | 99<br>Es<br>[252.1]<br>Einsteinium | 100<br>Fm<br>[257.1]<br>Fermium | 101<br>Md<br>[258.1]<br>Mendelevium | 102<br>No<br>[259.1]<br>Nobelium | 103<br>Lr<br>[262.1]<br>Lawrencium |
|---------------------------------|------------------------------|-----------------------------------|-----------------------------|----------------------------------|----------------------------------|----------------------------------|-------------------------------|----------------------------------|------------------------------------|------------------------------------|---------------------------------|-------------------------------------|----------------------------------|------------------------------------|

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.  
The atomic weights of Np and Tc are given for the isotopes <sup>237</sup>Np and <sup>99</sup>Tc.

**2003 HSC Notes from  
the Marking Centre  
Physics**

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# 2003 HSC NOTES FROM THE MARKING CENTRE

## PHYSICS

### Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It provides comments with regard to responses to the 2003 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2003 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

### General Comments

In 2003, approximately 9501 candidates attempted the Physics examination.

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections, should accumulate to a more comprehensive understanding than may be described in each section separately.

### Section I – Core

#### Part A – Multiple choice

| Question | Correct Response |
|----------|------------------|
| 1        | B                |
| 2        | A                |
| 3        | B                |
| 4        | A                |
| 5        | B                |
| 6        | B                |
| 7        | D                |
| 8        | C                |

| Question | Correct Response |
|----------|------------------|
| 9        | C                |
| 10       | A                |
| 11       | B                |
| 12       | D                |
| 13       | D                |
| 14       | D                |
| 15       | A                |
|          |                  |

## Part B

### General Comments

There is substantial evidence in responses to indicate that candidates are not following the instructions provided on the examination paper.

- (a) Answer the question in the spaces provided.  
A great number of responses contained irrelevant information, suggesting that answers had been rote-learned to address specific areas of the syllabus, then written down regardless of what the question was actually asking. Candidates should, where possible, make use of clearly labelled diagrams to supplement a concise response. The space provided should be used as a guide to the length of the response required. Additional (blank) spaces may be used to plan or draft responses. By providing additional information beyond the requirements of the question, candidates may disadvantage themselves by introducing contradictions.
- (b) Show all relevant working in questions involving calculations.  
An incorrect answer standing alone cannot score marks. Responses showing correct equations and substitutions but incorrect calculations may score some marks.

### Specific Comments

#### Question 16

On balance all parts of this question were well answered and presented few difficulties to the bulk of the candidature. A common error in part (b) was the placement of dependent and independent variables on the wrong axes.

#### Question 17

- (a) Many candidates wrongly substituted 9.8 for  $G$  and others used Newton's very similar equation for gravitational force.
- (b) Most candidates appreciated the equality of energies but then had trouble manipulating the equations to give  $v$ .
- (c) Many candidates had a very poor understanding of the physics behind launching a satellite. Even those who knew that most are launched towards the east, were not sure why.

#### Question 18

- (a) Most candidates did not correctly give two features of the aether model and instead gave two properties of aether.
- (b) Good responses consisted of a series of succinct, sequential steps, with a description of the anticipated result and using key phrases such as 'interferometer' and 'interference pattern'. Neat, labelled diagrams of the interferometer were a feature of many responses. Weaker responses illustrated poor understanding of the apparatus or procedure used by Michelson and Morley, or of the theoretical concepts involved.

### Question 19

The majority of responses included the substitution of correct values into the appropriate equation; however there were some errors in conversion of units and in recognising the correct length. Some candidates could not indicate the correct direction of the force on the wires.

### Question 20

Many candidates failed to recognise the significance and purpose of the copper wire in the second figure or did not interpret the question correctly. Better candidates, having carefully considered the information given, were able to give a succinct and logically sequential response to the question to gain full marks. First-hand practical knowledge seemed to have assisted these candidates. The majority of the candidates displayed some knowledge of the theoretical aspects of electromagnetic induction.

### Question 21

Many responses were longer than was indicated by the space provided.

- (a) Superior responses provided the cause and effect relationship between number of turns in the coils, voltage and current. Incorrect responses included power transmission.
- (b) The better responses linked the concept of changing magnetic flux to how a transformer worked, and identified the differences between AC and DC.

### Question 22

A very significant proportion of candidates misinterpreted this question, despite it being a direct quote from the syllabus. They read the question so that ‘generated current’ became ‘pre-existing current’ and so dealt with the motor effect. From statements written by the candidates it was obvious that the question had been misinterpreted rather than the correct answer not being known. Many candidates knew that there was a relationship between induced current in a conductor and change in flux although many confused a change in flux with magnetic field strength.

Many candidates had no concept of controlled variables. They understood this to mean variables that the experimenter varied, or to repeat the experiment. A large number of candidates ignored this section of the question or simply listed variables without indicating ‘how’ they were controlled.

A large number of candidates had no idea of how ammeters and voltmeters are connected in circuits and many believe that current is measured on a voltmeter.

A disappointingly large number of candidates put a DC source of current into their sketched circuit.

### Question 23

- (a) Many candidates simply described the Meissner effect without explaining the balance between the weight force and the force of magnetic repulsion resulting from the interaction of the fields of the induced current and hovering magnet.

- (b) Candidates knew a lot about conduction in metals and superconductors; however very few made comparisons between the two models. Very few responses consisted of tables to make comparisons, with the great majority of candidates simply listing features of both, and using no linking or comparative terms. Those responses including clearly labelled diagrams of electrons moving through both structures usually scored full marks.

#### Question 24

Many candidates were familiar with Thomson's experiment but a significant number were unable to outline both parts of it. Still fewer candidates were able to show how Thomson linked these two parts. Most candidates who used equations were able to use them to show the link between the two parts of Thomson's experiment.

A significant number of candidates confused Thomson's experiment with some other cathode ray experiment or with Millikan's oil drop experiment.

The better responses showed evidence of planning to use only the space provided. They included diagrams as an aid to their outline, although a significant number of these were confusing and misleading.

#### Question 25

- (a) Most candidates were able to use the data given to calculate the energy of the photon, although some confused wavelength with frequency.
- (b) Most candidates were able to cite the work function or the threshold frequency as an explanation for the observation described in the question, but many then went on to explain what would happen if the frequency was increased without explaining why it would happen in terms of photon energy.

Better responses contained reference to part (a) to provide information in a quantum explanation involving the energy of the photons.

#### Question 26

Many responses included some description of Einstein's contribution to Special Relativity and to Quantum Theory, but few successfully linked the contributions to how they changed the direction of scientific thought in the 20th century.

#### Question 27

The better responses connected the stimulus material with the question and made the appropriate link between centripetal force and the magnetic force. They then correctly identified mass dilation and described the appropriate change to the magnetic field.

Weaker responses made the link between centripetal force and magnetic force but then failed to take into account mass dilation or did not answer the question with respect to the magnetic field.

## Section II – Options

### Question 28 – Option Geophysics

In general candidates performed poorly on this question as it attracted many single student responses from large centres, where candidates apparently attempted the question without previously studying the option.

A few responses included good use of diagrams and figures.

- (a) Some candidates were unable to clarify ‘principal’ methods.
- (b) A minority of candidates displayed a good understanding of the topic.
- (c) Many candidates ignored the resources supplied with the question (picture and text), and gave a general discussion on remote sensing.
- (d) In the better responses, candidates demonstrated a clear understanding of the components of a geophone and could relate it to other basic physics principles.
- (e) Good responses linked physics theory to data collected in surveys and used it to demonstrate how this impacted on the greater understanding of Earth’s structure. In the weaker responses, candidates struggled to apply principles of physics to their answer.

### Question 29 – Option Medical Physics

Many candidates seemed to be confused and unable to adequately distinguish the five basic diagnostic procedures in the Medical Physics syllabus section.

A large number of candidates appeared to have general, not specific knowledge of medical imaging. It is important that candidates address the physics involved.

- (a) (i) The majority of candidates responded well, with a broad range of possible correct answers being accepted.
- (a) (ii) Better responses described both Bremsstrahlung and characteristic X-radiation. Most had some notion of braking radiation (Bremsstrahlung).
- (b) This question was well answered by many candidates. The best responses described the manner in which attenuation of gamma rays by tissue was used to pinpoint the location of the radiopharmaceutical concentration within the body. Candidates often confused CAT or Technetium-based gamma scans with PET.
- (c) (i) The better responses demonstrated the similarities and differences of these two scans. Poorer responses gave stand-alone descriptions of the two scanning techniques. A significant number confused PET scans with bone scans.
- (c) (ii) The best responses identified that a radiopharmaceutical is tagged with an isotope and so can target and subsequently identify parts of the body, eg bone cancers, through following metabolic pathways. A significant number of responses did not address the question being asked.
- (d) (i) This question was answered well.
- (d) (ii) Most candidates realised the difference in acoustic impedance was needed to answer the question correctly. Some candidates made incorrect calculations.

- (d) (iii) The few good responses linked the properties of ultrasound to the measurements made of variations in bone density. Poorer responses only stated general properties of ultrasound.
- (e) The better responses included information to support statements about electrons in CAT and PET imaging production and detection. Poorer responses described CAT and PET scans in general and compared the techniques.

### Question 30 – Option Astrophysics

- (a) Most candidates could define the resolution of a telescope, and were able to identify a means by which the resolution of a ground-based telescope could be improved. Poorer responses included the description of a different method, or indicated minimal knowledge of how their chosen method functioned.
- (b) Generally candidates performed well in this question but had difficulty manipulating the ‘ $\log d/10$ ’.
- (c) Responses indicated that candidates seemed confused by emission spectrum.
- (c) (i) Most candidates did not notice the red shift and described the bands.
- (c) (ii) (1) Many candidates described absorption correctly but did not relate it to a star.
- (c) (ii) (2) The many poor responses often incorrectly referred to black body radiation or described emission spectrum.
- (d) (i) Many candidates considered the cluster to be one star at the point it was labelled or assumed any star on the main sequence must be young.
- (d) (ii) (1) Many candidates thought the CNO cycle resulted in heavier elements being produced rather than carbon as a catalyst.
- (d) (ii) (2) Many candidates were able to give an evolutionary path but did not proceed to a corpse eg neutron star or black hole.
- (e) In the best responses, candidates not only specified how the light spectrum has been used to interpret certain types of information, they continued on to define impacts this information has had upon the modern scientific view of stars, nebulae, galaxies, etc. The better responses identified at least one impact, and then gave a value judgement as to its significance.

Poorer responses used the phrase ‘the visible spectrum of light’, without defining precisely how it was being used. In the majority of cases the candidates meant the use of spectral lines, but if these were not clearly identified, or at least described, the properties of stars depending upon the absorption (or emission) spectra were simply a jumble of facts.

### Question 31 – Option Quantum to Quarks

- (a) This part was generally well answered though a minority of candidates confused Thompson’s and Rutherford’s models of the atom.
- (b) This part was generally well answered.
- (c) The better responses linked the isotope production to its parent reaction. In the poorer responses, candidates did not know the processes involved in generating radioactive

isotopes, although they could explain the uses of the isotopes.

- (d) (i) Generally well answered. Some candidates performed calculations for the forces when it would have been easier to read them from the graph.
- (d) (ii) Some candidates did not interpret the question in sufficient depth. Their explanation of the importance of the strong nuclear force was not described adequately. Various groups of candidates mentioned gravitational and electrostatic forces and then talked of the repulsive force between the protons as if there was a third force.
- (e) Many candidates had a good understanding of the basic differences between controlled and uncontrolled nuclear reactions.

While basic nuclear reactor design was reasonably well described many candidates had a poor understanding of the role of a moderator in sustaining a controlled reaction in a conventional reactor.

### **Question 32 – Option Age of Silicon**

- (a) There was considerable confusion about the difference between digital and analogue operation.
- (b) Candidates who understood the question performed very well.
- (c) A significant number of candidates failed to notice the logarithmic scale
- (d) Some candidates commonly use symbols in their answers without defining the quantities the symbols represent, making their answers ambiguous. Most candidates were unfamiliar with the idea of voltage division in a series of resistors. Many candidates were not careful in reading the graph, and selected the nearest ‘whole-number’ along the curve, rather than making an appropriate approximation.
- (e) Essay answers were generally concise and well-written.

# Physics

## 2003 HSC Examination Mapping Grid

| Question | Marks | Content      | Syllabus outcomes |
|----------|-------|--------------|-------------------|
| 1        | 1     | 9.2.1        | H9                |
| 2        | 1     | 9.2.2        | H6                |
| 3        | 1     | 9.2.2, 9.2.3 | H6, H9            |
| 4        | 1     | 9.2.2        | H9                |
| 5        | 1     | 9.2.4        | H6                |
| 6        | 1     | 9.3.3        | H12               |
| 7        | 1     | 9.3.5        | H9                |
| 8        | 1     | 9.3.4        | H9                |
| 9        | 1     | 9.3.1        | H9                |
| 10       | 1     | 9.3.1        | H9                |
| 11       | 1     | 9.4.4        | H1                |
| 12       | 1     | 9.4.1        | H8, H13, H14      |
| 13       | 1     | 9.4.3        | H7                |
| 14       | 1     | 9.4.2        | H8                |
| 15       | 1     | 9.4.1        | H9                |
| 16 (a)   | 1     | 9.1          | H14               |
| 16 (b)   | 3     | 9.1          | H14               |
| 16 (c)   | 2     | 9.2.2        | H14, H9           |
| 17 (a)   | 1     | 9.2.1        | H9, H14           |
| 17 (b)   | 3     | 9.2.2        | H9, H14           |
| 17 (c)   | 2     | 9.2.2        | H9, H14           |
| 18 (a)   | 2     | 9.2.4        | H8                |
| 18 (b)   | 4     | 9.2.4        | H8, H1            |
| 19       | 3     | 9.3.1        | H9, H14           |
| 20       | 4     | 9.3.2        | H6, H9            |

| Question        | Marks | Content             | Syllabus outcomes |
|-----------------|-------|---------------------|-------------------|
| 21 (a)          | 3     | 9.3.4               | H7                |
| 21 (b)          | 2     | 9.3.4               | H7                |
| 22              | 5     | 9.3.2               | H9                |
| 23 (a)          | 3     | 9.3.2, 9.4.4        | H6, H7, H9        |
| 23 (b)          | 3     | 9.4.2               | H2, H9            |
| 24              | 4     | 9.4.4               | H9                |
| 25 (a)          | 2     | 9.4.2               | H8                |
| 25 (b)          | 3     | 9.4.2               | H8                |
| 26              | 6     | 9.2.4, 9.4.2        | H1                |
| 27              | 4     | 9.2.2, 9.2.4, 9.4.1 | H7                |
| 28 (a) (i)      | 1     | 9.5.1               | H10               |
| 28 (a) (ii)     | 2     | 9.5.5               | H3                |
| 28 (b)          | 3     | 9.5.4               | H6                |
| 28 (c) (i)      | 2     | 9.5.2               | H8, H10           |
| 28 (c) (ii)     | 3     | 9.5.2               | H8, H10, H14      |
| 28 (d) (i)      | 2     | 9.5.3               | H8, H13           |
| 28 (d) (ii) (1) | 2     | 9.5.3               | H8, H13           |
| 28 (d) (ii) (2) | 2     | 9.5.3               | H8, H13           |
| 28 (e)          | 8     | 9.2.3, 9.5.2        | H4, H8            |
| 29 (a) (i)      | 1     | 9.6.4               | H9                |
| 29 (a) (ii)     | 2     | 9.6.2               | H7                |
| 29 (b)          | 3     | 9.6.3               | H7                |
| 29 (c) (i)      | 3     | 9.6.3               | H7, H8, H10, H11  |
| 29 (c) (ii)     | 2     | 9.6.3               | H7, H11           |
| 29 (d) (i)      | 1     | 9.6.1               | H7, H8, H13       |
| 29 (d) (ii)     | 2     | 9.6.1               | H7, H8, H13       |
| 29 (d) (iii)    | 3     | 9.6.1               | H7, H8, H13       |
| 29 (e)          | 8     | 9.6.2, 9.6.3, 9.6.4 | H4, H7, H8        |

| Question        | Marks | Content           | Syllabus outcomes |
|-----------------|-------|-------------------|-------------------|
| 30 (a) (i)      | 1     | 9.7.1             | H8                |
| 30 (a) (ii)     | 2     | 9.7.1             | H8                |
| 30 (b)          | 3     | 9.7.4             | H8, H14           |
| 30 (c) (i)      | 1     | 9.7.3             | H8, H14           |
| 30 (c) (ii) (1) | 2     | 9.7.3             | H8, H14           |
| 30 (c) (ii) (2) | 2     | 9.7.3             | H10, H14          |
| 30 (d) (i)      | 1     | 9.7.6             | H9                |
| 30 (d) (ii) (1) | 2     | 9.7.6             | H7, H13           |
| 30 (d) (ii) (2) | 3     | 9.7.6             | H7                |
| 30 (e)          | 8     | 9.7.1, 3, 4, 5, 6 | H4, H8            |
| 31 (a) (i)      | 1     | 9.8.1             | H7                |
| 31 (a) (ii)     | 2     | 9.8.1             | H7                |
| 31 (b)          | 3     | 9.8.4             | H10               |
| 31 (c) (i)      | 2     | 9.8.3             | H7, H13           |
| 31 (c) (ii)     | 3     | 9.8.3, 9.8.4      | H7, H13           |
| 31 (d) (i)      | 2     | 9.8.3             | H6, H9, H13       |
| 31 (d) (ii)     | 2     | 9.8.3             | H9, H13           |
| 31 (d) (iii)    | 2     | 9.8.3             | H9, H13           |
| 31 (e)          | 8     | 9.8.3, 9.8.4      | H4, H7            |
| 32 (a) (i)      | 1     | 9.9.2             | H7                |
| 32 (a) (ii)     | 2     | 9.9.2             | H7                |
| 32 (b)          | 3     | 9.9.5             | H9                |
| 32 (c) (i)      | 3     | 9.9.7             | H9, H14           |
| 32 (c) (ii)     | 2     | 9.9.7             | H9, H14           |
| 32 (d) (i)      | 3     | 9.9.3, 9.9.2      | H7, H13           |
| 32 (d) (ii)     | 3     | 9.9.6             | H7, H13           |
| 32 (e)          | 8     | 9.9.3, 9.9.4      | H4, H7            |

## 2003 HSC Physics Marking Guidelines

### Question 16 (a)

*Outcomes assessed: H14*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                        | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"><li>Students can read the gradient of the line from the content of the box on the graph gradient = velocity</li></ul> OR <ul style="list-style-type: none"><li>Student correctly calculates gradient from data taken from the graph</li></ul> | 1     |

### Question 16 (b)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Mark correctly labelled axes (includes units)</li> <li>Straight line indicating a linear relationship between horizontal velocity and range and approaching the origin</li> <li>Point (1.85, 0.6) must be clearly identified as being on the line. (The point 1.85, 0.6) may be implied by numerical divisions on the axes)</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Mark correctly labelled axes (includes units)</li> </ul> AND <ul style="list-style-type: none"> <li>Straight line approaching origin or point (1.85, 0.6) identified</li> </ul> OR <ul style="list-style-type: none"> <li>Straight line approaching origin and point (1.85, 0.6) identified, no axes labelled</li> </ul>               | 2     |
| <ul style="list-style-type: none"> <li>Correctly labelled axes (units not required)</li> </ul> OR <ul style="list-style-type: none"> <li>Straight line approaching origin</li> </ul> OR <ul style="list-style-type: none"> <li>Point (1.85, 0.6) identified</li> </ul>                                                                                                        | 1     |

### Question 16 (c)

Outcomes assessed: H14, H9

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Range on the planet is longer because of the acceleration due to gravity (<math>g</math>) on the planet is smaller</li> </ul> OR <ul style="list-style-type: none"> <li>Since <math>g</math> is smaller on the planet the time of flight is longer</li> </ul> OR <ul style="list-style-type: none"> <li>Longer time of flight means that the ball travels a longer horizontal distance before it reaches the ground</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Longer because gravity on the planet is smaller</li> </ul> OR <ul style="list-style-type: none"> <li>Longer because time of flight is longer</li> </ul>                                                                                                                                                                                                                                                                        | 1     |

**Question 17 (a)**

*Outcomes assessed: H14, H9*

**MARKING GUIDELINES**

| Criteria                                                                                                  | Marks |
|-----------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identify correct equation</li> <li>Correct substitution</li> </ul> | 1     |

**Question 17 (b)**

*Outcomes assessed: H14, H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Relationship given in question hence equating <math> E_p  = E_k</math> is given</li> </ul> AND <ul style="list-style-type: none"> <li>Correct expression for <math>v</math>, or inferred</li> </ul> AND <ul style="list-style-type: none"> <li>Correct substitution, note that the answer (even if wrong) from (a) can be substituted</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Relationship given in question hence equating <math> E_p  = E_k</math> is given</li> </ul> AND <ul style="list-style-type: none"> <li>Mark given for correct expression for <math>v</math>, or inferred</li> </ul>                                                                                                                               | 2     |
| <ul style="list-style-type: none"> <li><math> E_p  = E_k</math></li> </ul>                                                                                                                                                                                                                                                                                                              | 1     |

**Question 17(c)**

*Outcomes assessed: H9, H14*

**MARKING GUIDELINES**

| Criteria                                                                                              | Marks |
|-------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>An issue and relationship to direction of launch</li> </ul>    | 2     |
| <ul style="list-style-type: none"> <li>An issue such as assistance to motion or fuel usage</li> </ul> | 1     |

### Question 18 (a)

Outcomes assessed: H8

#### MARKING GUIDELINES

| Criteria                                                                                                                       | Marks |
|--------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Aether is medium for propagation of light/EM radiation and one other feature</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Aether is medium for propagation of light/EM radiation OR another feature</li> </ul>    | 1     |

### Question 18 (b)

Outcomes assessed: H8, H1

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Constructed an interferometer which splits a single beam of light and sent them on perpendicular paths and reflected them back</li> <li>Observed an interference pattern when light beams returned</li> <li>The arms of apparatus rotated through <math>90^\circ</math> and experiment repeated</li> <li>It was expected that the two interference patterns would be different since the earth's passage through the aether would affect the velocity of light</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Includes three points from above</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               | 3     |
| <ul style="list-style-type: none"> <li>Includes two points from above</li> </ul> OR <ul style="list-style-type: none"> <li>Correct diagram of interferometer</li> </ul>                                                                                                                                                                                                                                                                                                                                          | 2     |
| <ul style="list-style-type: none"> <li>Used an interferometer or description of interferometer</li> </ul> OR <ul style="list-style-type: none"> <li>Use of interference patterns</li> </ul> OR <ul style="list-style-type: none"> <li>Any other one correct point from above</li> </ul>                                                                                                                                                                                                                          | 1     |

### Question 19

Outcomes assessed: H9, H14

#### MARKING GUIDELINES

| Criteria                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct calculation</li> <li>Correct direction</li> </ul>                | 3     |
| <ul style="list-style-type: none"> <li>Correct formula and correct substitution or correct direction</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Correct formula identified</li> </ul>                                    | 1     |

## Question 20

Outcomes assessed: H6, H9

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Fully explains both investigations which could include <ul style="list-style-type: none"> <li>Investigation 1: Since the solenoid is not in a circuit, no current can be induced by the magnetic field. No current <math>\rightarrow</math> no induced magnetic field <math>\rightarrow</math> no motion.</li> <li>Investigation 2: The circuit is closed, so magnet induces current in solenoid. Lenz's law says the current will produce a magnetic field opposing the changing field that induced it. So direction of induced field is opposite to field of magnet. They repel, and solenoid moves to the LEFT. Includes both a description of Lenz's law and gets direction of motion correct</li> </ul> </li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Explains both investigations</li> <li>Mentions Lenz's law OR gets direction of motion correct</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>Briefly explains both investigations</li> </ul> OR <ul style="list-style-type: none"> <li>Full explanation of either investigation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <ul style="list-style-type: none"> <li>Solenoid moves in investigation 2 but not in investigation 1</li> </ul> OR <ul style="list-style-type: none"> <li>Brief explanation of investigation 1 (no circuit, no motion)</li> </ul> OR <ul style="list-style-type: none"> <li>Brief explanation of investigation 2</li> </ul> OR <ul style="list-style-type: none"> <li>EMF induced in coil 1 because of the motion of the magnet</li> <li>OR</li> <li>No current produced in coil 1 but is in coil 2</li> </ul>                                                                                                                                                                                                                                                   | 1     |

**Question 21 (a)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                         | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Conservation of energy requires <math>E = V_p I_p t = V_s I_s t</math> hence <math>V_p I_p = V_s I_s</math></li> <li>If stepdown <math>V_p &gt; V_s</math></li> <li>which means that <math>I_p &lt; I_s</math></li> </ul> | 3     |
| Any two of the above                                                                                                                                                                                                                                             | 2     |
| Any one of the above                                                                                                                                                                                                                                             | 1     |

**Question 21 (b)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>A changing magnetic flux through the secondary coil is needed to induce a voltage/current</li> </ul> AND <ul style="list-style-type: none"> <li>An AC current in the primary coil produces a changing flux</li> </ul> AND <ul style="list-style-type: none"> <li>A DC current in the primary coil produces a constant magnetic flux</li> </ul> | 2     |
| One of the above points                                                                                                                                                                                                                                                                                                                                                               | 1     |

## Question 22

Outcomes assessed: H9

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Labelled diagram connected to CRO or meter and magnet nearby</li> </ul> AND <ul style="list-style-type: none"> <li>Description of how magnetic field varied</li> </ul> AND <ul style="list-style-type: none"> <li>Description of how two variables are controlled</li> </ul>                                                                     | 5     |
| <ul style="list-style-type: none"> <li>Failed to adequately describe either magnetic field variation or control of variables or draw complete diagram or to completely label the diagram</li> </ul>                                                                                                                                                                                     | 4     |
| <ul style="list-style-type: none"> <li>Failed to describe both magnetic field variation and control of variable</li> </ul> OR <ul style="list-style-type: none"> <li>Failed to provide adequate description of experimental setup or a fully labelled diagram</li> </ul> OR <ul style="list-style-type: none"> <li>Failed to provide labelled diagram or experimental set-up</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Inadequately labelled</li> </ul> OR <ul style="list-style-type: none"> <li>Incomplete diagram</li> </ul> AND <ul style="list-style-type: none"> <li>Description of how magnetic field varied</li> </ul> OR <ul style="list-style-type: none"> <li>How two variables are controlled</li> </ul>                                                    | 2     |
| <ul style="list-style-type: none"> <li>A magnet was pushed into coil connected to a meter</li> </ul> OR <ul style="list-style-type: none"> <li>Control of variables mentioned</li> </ul> OR <ul style="list-style-type: none"> <li>Mention of variation of magnetic field</li> </ul>                                                                                                    | 1     |

**Question 23 (a)**

*Outcomes assessed: H6, H7, H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>The magnetic field of the superconductor interacts with/excludes/opposes the field of the magnet and when in the superconducting state, current in the superconductor produces a magnetic field</li> </ul> AND <ul style="list-style-type: none"> <li>Force due to magnetic effect = force due to gravity in the opposite direction</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Any two of the above</li> </ul>                                                                                                                                                                                                                                                                                                                | 2     |
| <ul style="list-style-type: none"> <li>Any one of the above</li> </ul>                                                                                                                                                                                                                                                                                                                | 1     |

**Question 23 (b)**

*Outcomes assessed: H2, H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                   | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Comparison of one similarity and two differences</li> </ul> OR <ul style="list-style-type: none"> <li>Three similarities</li> </ul> OR <ul style="list-style-type: none"> <li>Three differences</li> </ul> OR <ul style="list-style-type: none"> <li>One difference and two similarities</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Comparison with one similarity and one difference or two differences</li> </ul> OR <ul style="list-style-type: none"> <li>Two similarities</li> </ul>                                                                                                                                               | 2     |
| <ul style="list-style-type: none"> <li>One similarity or one difference</li> </ul>                                                                                                                                                                                                                                                         | 1     |

### Question 24

Outcomes assessed: H9

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                    | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Clear outline of relationship between electric and magnetic fields and radius of curvature/in order to determine <math>\frac{q}{m}</math> (diagrams and equations may be used to support answer)</li> </ul>                          | 4     |
| <ul style="list-style-type: none"> <li>Outline of experimental set-up</li> </ul> AND <ul style="list-style-type: none"> <li>Uses electric and magnetic fields to balance forces on electron beam</li> <li>And centripetal force/radius of curvature/displacement</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Uses electric and magnetic fields to balance forces on electron beam</li> <li>And centripetal force/radius of curvature/displacement</li> </ul> OR <ul style="list-style-type: none"> <li>Outline of experimental set-up</li> </ul>  | 2     |
| <ul style="list-style-type: none"> <li>Uses electric and magnetic fields to balance forces on electron beam</li> </ul> OR <ul style="list-style-type: none"> <li>Uses only one field and notes position</li> </ul>                                                          | 1     |

### Question 25 (a)

Outcomes assessed: H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                         | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li><math>E = 1.28 \times 10^{-19} \text{ J}</math></li> </ul> OR <ul style="list-style-type: none"> <li>With correct substitution</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li><math>E = \frac{hc}{\lambda}</math></li> </ul> OR <ul style="list-style-type: none"> <li><math>f</math> correctly calculated</li> </ul>   | 1     |

### Question 25 (b)

Outcomes assessed: H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| All 3 of the following bullet points: <ul style="list-style-type: none"> <li>The photon needs to have a minimum energy/frequency to excite an electron to leave the surface</li> <li>For these photons the energy is too low</li> </ul> OR <ul style="list-style-type: none"> <li>An increase in energy is required</li> <li>Increasing the intensity (number of photons) does not increase the energy of the photons</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Any two bullet points in 'full answer'</li> </ul>                                                                                                                                                                                                                                                                                                                                         | 2     |
| <ul style="list-style-type: none"> <li>Any one bullet point in 'full answer'</li> </ul>                                                                                                                                                                                                                                                                                                                                          | 1     |

### Question 26

Outcomes assessed: H1

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                     | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Describes a contribution to each field and how each contribution changed scientific thinking in the 20<sup>th</sup> Century</li> </ul>                                                | 6     |
| <ul style="list-style-type: none"> <li>Description of a contribution to each field and how at least one contribution changed scientific thinking in the 20<sup>th</sup> Century</li> </ul>                                   | 5     |
| <ul style="list-style-type: none"> <li>Describes a contribution to each field</li> <li>OR</li> <li>Describes a contribution to one field and how the contribution changed scientific thinking in the 20th century</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Description of contribution to one field and mentions a point from another</li> </ul>                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>Any relevant points from each field</li> </ul> OR <ul style="list-style-type: none"> <li>Description of a contribution to one field</li> </ul>                                        | 2     |
| <ul style="list-style-type: none"> <li>Any one relevant point from either field</li> </ul>                                                                                                                                   | 1     |

### Question 27

Outcomes assessed: H7

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Clear description of why <math>B</math> must increase as <math>v</math> increases, and that as <math>v</math> approaches <math>c</math>, relativistic mass increases (with or without mass dilation equation)</li> </ul> <p>Using <math>F = \frac{mv^2}{r} = qvB</math></p>                                                                                                                                                                                                                                 | 4     |
| <ul style="list-style-type: none"> <li>Relates <math>F = qvB</math> to <math>F = \frac{mv^2}{r}</math> and states that <math>B</math> increases</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Relates <math>F = qvB</math> to <math>F = \frac{mv^2}{r}</math> and states the relativistic effects on the mass</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>States <math>F = qvB</math> <u>or</u> <math>F = \frac{mv^2}{r}</math> and states the relativistic effect on mass and <math>B</math> gets larger</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Same as three (3) marks but does not relate the equations</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>States two points from (1) mark</li> </ul>                                                                                                                                                                                                                                                                                                                                            | 2     |
| <ul style="list-style-type: none"> <li>As electron goes faster, mag field must be larger</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Accounts for relativistic mass/time</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Faster the electron goes greater force is required</li> </ul>                                                                                                                                                                                                                                 | 1     |

### Question 28 (a) (i)

Outcomes assessed: H10

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                               | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Must name any three of the following: seismic, gravitational, magnetic, palaeomagnetic, electrical, electromagnetic, radiometric, geothermal methods</li> </ul> | 1     |

**Question 28 (a) (ii)**

*Outcomes assessed: H3*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                 | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Compare possible tests against records of known geophysical activity or an answer covering similar ideas</li> </ul> AND one of: <ul style="list-style-type: none"> <li>Worldwide network of monitoring stations</li> <li>Variety of geophysical data to record possible nuclear tests,</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Need to mention ONE of the above points</li> </ul>                                                                                                                                                                                                                                                | 1     |

**Question 28 (b)**

*Outcomes assessed: H6*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                  | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Outlines three (or more) main pieces of evidence linking to plate tectonics</li> </ul> OR <ul style="list-style-type: none"> <li>Discusses two in depth</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Outlines two pieces of evidence, linked to plate tectonics</li> </ul> OR <ul style="list-style-type: none"> <li>Discusses one in depth</li> </ul>                  | 2     |
| <ul style="list-style-type: none"> <li>Outlines one piece of evidence, linked to plate tectonics</li> </ul>                                                                                               | 1     |

**Question 28 (c) (i)**

*Outcomes assessed: H8,H10*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                              | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Describe that different surfaces absorb/reflect different amounts of radiation, and that observing at different frequencies (or wavelengths) give more information about the object</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Describe ONE point from above</li> </ul>                                                                                                                                                       | 1     |

**Question 28 (c) (ii)**

*Outcomes assessed: H8, H10, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                   | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Spread of fire – smoke plume and blackened ground</li> </ul> AND <ul style="list-style-type: none"> <li>Changes in images over time to indicate regrowth</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Spread of fire – either of two points</li> </ul> AND <ul style="list-style-type: none"> <li>Changes in vegetation shown by different images</li> </ul>              | 2     |
| <ul style="list-style-type: none"> <li>Any one from above</li> </ul>                                                                                                                                       | 1     |

**Question 28 (d) (i)**

*Outcomes assessed: H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                         | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Magnet or spring, coil and both: relative motion produces current, and seismic waves cause relative motion, causes current in coil</li> </ul> OR <ul style="list-style-type: none"> <li>Similar details for other type geophone</li> </ul> OR <ul style="list-style-type: none"> <li>Suitable labelled diagram</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Magnet or spring, coil and relative motion produces current in coil</li> </ul> OR <ul style="list-style-type: none"> <li>Similar detail for type of geophone</li> </ul> OR <ul style="list-style-type: none"> <li>Similar diagram</li> </ul>                                                                              | 1     |

**Question 28 (d) (ii) (1)**

*Outcomes assessed: H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Refracted wave travels faster than direct wave</li> </ul> AND <ul style="list-style-type: none"> <li>Refracted wave initially arrives at geophones after the direct wave as it has further to travel, but overtakes it</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Refracted wave travels faster than direct wave</li> </ul>                                                                                                                                                                         | 1     |

**Question 28 (d) (ii) (2)**

*Outcomes assessed: H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct calculation</li> </ul>                                                                                                    | 2     |
| <ul style="list-style-type: none"> <li>For incorrect use of slope (time/distance)</li> </ul> OR <ul style="list-style-type: none"> <li>Incorrect substitution</li> </ul> | 1     |

### Question 28 (e)

Outcomes assessed: H4, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Well constructed, extensive answer on both the application of Newton's theory to geophysics, and the understanding of Earth's structure</li> <li>Should include <ul style="list-style-type: none"> <li>Statement of Newton's theory</li> <li>Geophysicists measure local gravitational force/acceleration through surveys with gravimeters (and/or satellite motion)</li> <li>Gravitational anomaly profiles give information about structure of Earth (density of material underground)</li> <li>Can get info about hidden structure not easily observed by other methods</li> </ul> </li> </ul> | 7–8   |
| <ul style="list-style-type: none"> <li>Sound knowledge outline of application of Newton's theory to geophysics and in discussion of how it leads to understanding of structure Earth</li> <li>Should include: <ul style="list-style-type: none"> <li>Geophysicists measure local gravity</li> <li>Gives info about density (structure) of Earth nearby</li> <li>Gravitational anomaly profiles shown landforms, structure etc</li> </ul> </li> </ul>                                                                                                                                                                                     | 5–6   |
| <ul style="list-style-type: none"> <li>Limited answer to both application of Newton's theory and understanding of Earth's structure</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>More extensive description of either application of Newton's theory or use in understanding Earth's structure</li> </ul>                                                                                                                                                                                                                                                                                                                 | 3–4   |
| <ul style="list-style-type: none"> <li>Limited outline of application of gravity to geophysics</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Limited discussion of how it leads to understanding of Earth's structure</li> <li>Any of the following could be included: <ul style="list-style-type: none"> <li>Local measurement of gravity</li> <li>Gravity shows hidden structure</li> <li>Local changes in gravity show structure</li> <li>Gravity depends on density of material</li> </ul> </li> </ul>                                                                                                                 | 1–2   |

**Question 29 (a) (i)**

*Outcomes assessed: H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                           | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Hydrogen nucleus behaves like a small magnet, magnetic property</li> <li>OR</li> <li>Hydrogen nuclei have a spin</li> </ul> | 1     |

**Question 29 (a) (ii)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                 | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Conversion of electrons kinetic energy to X-rays</li> <li>Excitation and subsequent emission electrons</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>One of the points above</li> </ul>                                                                                | 1     |

**Question 29 (b)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Radio-pharmaceutical produces positrons</li> <li>Pairs of <math>\gamma</math>-rays are produced by electron/positron annihilation</li> <li>Detectors paired on opposite sides of body detect gamma rays</li> <li>The attenuation of which depends on the depth of tissue the <math>\gamma</math>-rays have to pass through to reach the detectors</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Gamma rays are produced by electron/positron annihilation</li> <li>Pairs of gamma rays are detected on opposite sides of the body</li> </ul>                                                                                                                                                                                                                 | 2     |
| <ul style="list-style-type: none"> <li>One of the above</li> </ul>                                                                                                                                                                                                                                                                                                                                  | 1     |

**Question 29 (c) (i)**

*Outcomes assessed: H7, H8, H10, H11*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                             | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Compare location and nature of radiation source and production of image (different brightness due to: brightness of X-rays for X-ray image, different concentration of <math>\gamma</math> source in organs for bone scan)</li> </ul>                                                                         | 3     |
| <ul style="list-style-type: none"> <li>Must mention both location of radiation source and brightness of X-rays/concentration of <math>\gamma</math> source, and must compare one of these points</li> </ul>                                                                                                                                          | 2     |
| <ul style="list-style-type: none"> <li>Basic comparison (X-ray source outside body <math>\gamma</math> source inside body) or (simple statement about X-ray brightness vs <math>\gamma</math> source concentration)</li> </ul> OR <ul style="list-style-type: none"> <li>Good description of how radiation is used in X-ray or bone scans</li> </ul> | 1     |

**Question 29 (c) (ii)**

*Outcomes assessed: H7, H11*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                               | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Radio-pharmaceutical targets the bone due to selective uptake</li> <li>Hence the bone scan detects function (or physiology) due to the metabolic uptake of the radio-pharmaceutical rather than anatomy as do X-rays</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Radioisotopes are tagged to a molecule that targets/binds to the bone</li> </ul> OR <ul style="list-style-type: none"> <li>Gamma radiation shows sites of detailed structure change where X-rays may not</li> </ul>             | 1     |

**Question 29 (d) (i)**

*Outcomes assessed: H7, H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct substitution into formula</li> </ul> | 1     |

**Question 29 (d) (ii)**

*Outcomes assessed: H7, H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                         | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Identification of fat</li> <li>• Because of its greatest difference in acoustic impedance compared to kidney</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>• One of the above</li> </ul>                                                                                             | 1     |

**Question 29 (d) (iii)**

*Outcomes assessed: H7, H8, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Ultrasound can be passed through the (heel) bone from one side to the other (emitter/detector on opposite sides)</li> <li>• Variations in bone density result in the bone having a different acoustic impedance</li> <li>• Reflection from healthy bone/flesh boundary is different from reflection from abnormal bone/flesh boundary, therefore transmitted intensity is different</li> <li>• The speed of ultrasound is different in bones of different density</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>• Two of the points above</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| <ul style="list-style-type: none"> <li>• One of the points above</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1     |

### Question 29 (e)

Outcomes assessed: H4, H7, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Answer demonstrates extensive knowledge of the properties of electrons and the ability to control their behaviour and relates these to medical technologies including CAT and PET imaging technologies;</li> <li>Points, which should link charge of electrons to their control in these technologies, could include <ul style="list-style-type: none"> <li>Electrons accelerated in an X-ray tube to produce X-rays</li> <li>In PET, electrons controlled through their interaction with positrons to produce <math>\gamma</math> rays through particle/antiparticle annihilation (or matter/antimatter)</li> <li>Any of the technologies used involves electronic circuits in which electrons are controlled (by voltages/microunits) OR imaging technologies use cathodes ray tubes as a display device, which require electron control by electric and magnetic fields</li> <li>The electron has a negative charge and hence it can be deflected by electric and magnetic fields</li> </ul> </li> </ul> | 7–8   |
| <ul style="list-style-type: none"> <li>Answer demonstrates substantial knowledge of the properties of electrons and the ability to control them and relates these to both PET and CAT imaging technology</li> </ul> OR <ul style="list-style-type: none"> <li>Answer demonstrates substantial knowledge of the properties of electrons and the ability to control them and relates these to PET or CAT and at least one other medical technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5–6   |
| <ul style="list-style-type: none"> <li>Description of at least one property of electrons or the ability to control electrons and relates this to PET and/or CAT imaging technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2–4   |
| <ul style="list-style-type: none"> <li>One property of electrons</li> </ul> OR <ul style="list-style-type: none"> <li>A way electrons can be controlled</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |

### Question 30 (a) (i)

Outcomes assessed: H8

#### MARKING GUIDELINES

| Criteria                                                                                                                    | Marks |
|-----------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Definition showing understanding of distinguishing fine detail/separation</li> </ul> | 1     |

**Question 30 (a) (ii)**

*Outcomes assessed: H8*

**MARKING GUIDELINES**

| Criteria                                                                                                    | Marks |
|-------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>One method named</li> <li>Brief description of the method</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Method named only</li> </ul>                                         | 1     |

**Question 30 (b)**

*Outcomes assessed: H8, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct identification of <math>M</math> and <math>m</math></li> <li>Correct substitution into correct formula</li> </ul>                        | 3     |
| <ul style="list-style-type: none"> <li>Correct identification of <math>M</math> OR <math>m</math></li> <li>Correct substitution using values chosen into the correct formula</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Correct identification of either <math>M</math> OR <math>m</math></li> </ul>                                                                     | 1     |

**Question 30 (c) (i)**

*Outcomes assessed: H8, H14*

**MARKING GUIDELINES**

| Criteria                                                                                | Marks |
|-----------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies that redshift has occurred</li> </ul> | 1     |

**Question 30 (c) (ii) (1)**

*Outcomes assessed: H8, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Hotter gases in interior produce spectrum and cooler outer gases absorb radiation resulting in dark lines</li> </ul>                                                                                                                                                                       | 2     |
| <ul style="list-style-type: none"> <li>Radiation absorbed resulting in dark lines</li> </ul> OR <ul style="list-style-type: none"> <li>Spectrum produced by hot interior gases or core</li> </ul> OR <ul style="list-style-type: none"> <li>Describes how an absorption spectrum is produced but not related to a star</li> </ul> | 1     |

**Question 30 (c) (ii) (2)**

*Outcomes assessed: H10, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                  | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Presence of spectral lines associated with hotter or cooler temperatures</li> <li>• Specific example of an element/spectrum that demonstrates this</li> </ul>    | 2     |
| <ul style="list-style-type: none"> <li>• Specific example only</li> </ul> OR <ul style="list-style-type: none"> <li>• Presence of spectral lines associated with hotter or cooler temperatures</li> </ul> | 1     |

**Question 30 (d) (i)**

*Outcomes assessed: H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                                               | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• High turn off point</li> </ul> OR <ul style="list-style-type: none"> <li>• Still has B class stars</li> </ul> | 1     |

**Question 30 (d) (ii) (1)**

*Outcomes assessed: H13, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                             | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Star <i>X</i> – CN, C-N-O cycle dominates</li> <li>• Star <i>Z</i> – proton-proton chain is main process</li> </ul>                         | 2     |
| <ul style="list-style-type: none"> <li>• Only information on one star given or only information on one star correct</li> <li>• OR</li> <li>• Faster reactions in <i>X</i></li> </ul> | 1     |

**Question 30 (d) (ii) (2)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                          | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Distinction between mass left after becomes supernova to give two pathways</li> </ul>                                                                                                                                                                      | 3     |
| <ul style="list-style-type: none"> <li>Short life on main sequence, diagram without masses of cores</li> </ul> OR <ul style="list-style-type: none"> <li>Two paths but no core mass identified</li> </ul> OR <ul style="list-style-type: none"> <li>One path with core mass identified</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Diagram only without masses</li> </ul> OR <ul style="list-style-type: none"> <li>One path with no mass</li> </ul>                                                                                                                                          | 1     |

**Question 30 (e)**

*Outcomes assessed: H4, H8*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Extensive knowledge of information obtained from light</li> <li>At least three different ideas that have learnt about celestial objects from studying them using light spectrum</li> </ul> | 7–8   |
| <ul style="list-style-type: none"> <li>Thorough knowledge of information obtained from light spectrum</li> <li>At least three different ideas considered</li> </ul>                                                               | 5–6   |
| <ul style="list-style-type: none"> <li>Sound knowledge of information obtained from light spectrum</li> <li>At least two different ideas considered</li> </ul>                                                                    | 3–4   |
| <ul style="list-style-type: none"> <li>Basic knowledge of information obtained from light spectrum</li> <li>At least two different ideas considered</li> </ul>                                                                    | 1–2   |

**Question 31 (a) (i)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                    | Marks |
|-----------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Electrons orbit a nucleus</li> </ul> | 1     |

**Question 31 (a) (ii)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                   | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Electrons orbit the nucleus/instable orbits at discrete radii OR particular energy levels OR quantised energy levels</li> </ul> AND | 2     |
| <ul style="list-style-type: none"> <li>Electrons can move between levels by absorbing or emitting energy</li> </ul>                                                        |       |
| <ul style="list-style-type: none"> <li>One of the points listed above</li> </ul>                                                                                           | 1     |

**Question 31 (b)**

*Outcomes assessed: H10*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Protons have + 1 charge, while neutrons have 0 charge</li> <li>Protons are <i>uud</i></li> <li>Neutrons are <i>udd</i></li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Two point above</li> </ul>                                                                                                        | 2     |
| <ul style="list-style-type: none"> <li>One point above</li> </ul>                                                                                                        | 1     |

**Question 31 (c) (i)**

*Outcomes assessed: H13, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Reaction 2 because part of radioactive decay series</li> </ul> OR <ul style="list-style-type: none"> <li>U-238 is naturally occurring</li> </ul> OR <ul style="list-style-type: none"> <li>No energy has been put in</li> </ul> OR <ul style="list-style-type: none"> <li>No neutrons are added to it</li> </ul> OR <ul style="list-style-type: none"> <li>Reaction 3 — geological evidence in Africa</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Reaction 2</li> </ul> OR <ul style="list-style-type: none"> <li>Reaction 3</li> </ul>                                                                                                                                                                                                                                                                                                                            | 1     |

**Question 31 (c) (ii)**

*Outcomes assessed: H13, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Reaction 3</li> <li>Used nuclear reactors</li> <li>For power or weapons or radioisotope production or electrical energy production</li> <li>Plus details of named application</li> </ul> OR <ul style="list-style-type: none"> <li>Reaction 1</li> <li>Artificial transmutation</li> <li>Production of radioisotopes OR</li> <li>Production of plutonium or fast breeder reaction</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Correctly names reaction 1 or 3</li> <li>Identifies one application</li> </ul> OR <ul style="list-style-type: none"> <li>Describes a feature</li> </ul>                                                                                                                                                                                                                                      | 2     |
| <ul style="list-style-type: none"> <li>Names one reaction (1 or 3)</li> </ul>                                                                                                                                                                                                                                                                                                                                                       | 1     |

**Question 31 (d) (i)**

*Outcomes assessed: H6, H9, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                   | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li><math>F_G = -2 \times 10^{-34} \text{ N} (\pm 0.1 \times 10^{-34} \text{ N})</math> and <math>F_E = 230 \text{ N}</math></li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>One result</li> </ul>                                                                                                               | 1     |

**Question 31 (d) (ii)**

*Outcomes assessed: H9, H13*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>The electrostatic force is greater than the gravitational force</li> <li>Therefore the electrostatic force would blow the nucleus apart Both points below</li> </ul> OR <ul style="list-style-type: none"> <li>The strong nuclear force is needed to overcome the electrostatic force</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>One of the points above</li> </ul>                                                                                                                                                                                                                                                               | 1     |

**Question 31 (d) (iii)***Outcomes assessed: H9, H13***MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Any two of the following: <ul style="list-style-type: none"><li>• The strong nuclear force holds all nuclear particles together, whether charged or uncharged</li><li>• It is much stronger than the electrostatic force</li><li>• Is attractive only over a very small distance (<math>\sim 10^{-15}</math> m)</li><li>• Becomes repulsive at small distances — less than the diameter of a nucleon</li></ul> | 2     |
| <ul style="list-style-type: none"><li>• Any one of the points above</li></ul>                                                                                                                                                                                                                                                                                                                                  | 1     |

### Question 31 (e)

Outcomes assessed: H4, H7

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Well constructed extensive answer describing the requirements for a controlled nuclear chain reaction</li> </ul> AND <ul style="list-style-type: none"> <li>Well constructed extensive answer describing the requirements for an uncontrolled nuclear reaction</li> <li>Must show linkage between factors involved in controlled and uncontrolled reactions</li> </ul>                                                                                                                                                        | 7–8   |
| <ul style="list-style-type: none"> <li>Sound answer describing the requirements of a controlled nuclear chain reaction in a fission reactor</li> </ul> AND <ul style="list-style-type: none"> <li>Sound answer describing requirements for an uncontrolled nuclear reaction</li> <li>No linkages between factors involved in the two types of reactions</li> </ul>                                                                                                                                                                                                   | 5–6   |
| <ul style="list-style-type: none"> <li>Sound answer which describes the requirements for a controlled nuclear chain reaction</li> </ul> AND <ul style="list-style-type: none"> <li>Sound answer which describes the requirements for an uncontrolled chain reaction</li> </ul> OR <ul style="list-style-type: none"> <li>A limited answer describing the requirements of a controlled nuclear chain reaction</li> </ul> AND <ul style="list-style-type: none"> <li>A limited answer describing the requirements of an uncontrolled nuclear chain reaction</li> </ul> | 3–4   |
| <ul style="list-style-type: none"> <li>Limited answer which describes</li> <li>The control of the reaction</li> </ul> OR <ul style="list-style-type: none"> <li>Compares a similarity or difference</li> </ul>                                                                                                                                                                                                                                                                                                                                                       | 1–2   |

### Question 32 (a) (i)

Outcomes assessed: H7

#### MARKING GUIDELINES

| Criteria                                                                       | Marks |
|--------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>One correct example for each</li> </ul> | 1     |

**Question 32 (a) (ii)**

*Outcomes assessed: H7*

**MARKING GUIDELINES**

| Criteria                          | Marks |
|-----------------------------------|-------|
| • Two correctly identified graphs | 2     |
| • One is correct but not both     | 1     |

**Question 32 (b)**

*Outcomes assessed: H9*

**MARKING GUIDELINES**

| Criteria                                                                                                                     | Marks |
|------------------------------------------------------------------------------------------------------------------------------|-------|
| • Each of the steps listed below<br>( $P = \text{not } A$ , $Q = P \text{ or } B$ , $R = Q \text{ and } A$ ) is worth 1 mark | 3     |
| • Two of ( $P = \text{not } A$ , $Q = P \text{ or } B$ , $R = Q \text{ and } A$ ) correct                                    | 2     |
| • One of ( $P = \text{not } A$ , $Q = P \text{ or } B$ , $R = Q \text{ and } A$ )                                            | 1     |

**Question 32 (c) (i)**

*Outcomes assessed: H9, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                      | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Computers become faster</li> </ul> AND <ul style="list-style-type: none"> <li>Density increase so size of circuit elements decrease</li> </ul> OR <ul style="list-style-type: none"> <li>Density increases (by factor of <math>10^5</math>)</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Computers become faster</li> </ul> AND <ul style="list-style-type: none"> <li>Density increases</li> </ul> OR <ul style="list-style-type: none"> <li>Size of circuit elements decrease</li> </ul>                                                      | 2     |
| <ul style="list-style-type: none"> <li>Computers become faster</li> </ul> OR <ul style="list-style-type: none"> <li>Density increase</li> </ul> OR <ul style="list-style-type: none"> <li>Size of circuit elements decrease</li> </ul>                                                        | 1     |

**Question 32 (c) (ii)**

*Outcomes assessed: H9, H14*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                       | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Graph is not valid at 2060 (1 mark)</li> </ul> AND one of: <ul style="list-style-type: none"> <li>reduction in size</li> <li>more components – heat dissipation</li> <li>connection problems</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Quantum effects may cause changes in way computers operate</li> </ul>                                                                                                                                   |       |
| <ul style="list-style-type: none"> <li>1 of the above points</li> </ul>                                                                                                                                                                        | 1     |

**Question 32 (d) (i)**

*Outcomes assessed: H13, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                       | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Resistance at <math>15^{\circ}\text{C} = 1400\Omega</math> (1 mark)</li> <li>Voltage divider algebra (1 mark)</li> <li>Final answer including units (1 mark)</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Any 2 of above points</li> </ul>                                                                                                                                        | 2     |
| <ul style="list-style-type: none"> <li>Any 1 of above points</li> </ul>                                                                                                                                        | 1     |

**Question 32 (d) (ii)**

*Outcomes assessed: H13, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                            | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Gain equation <math>\frac{V_{\text{coil}}}{V_A} = \frac{-R}{10^5}</math></li> <li>Relationship between gain equations correct</li> <li>Correct substitution</li> </ul>                                                       | 3     |
| <ul style="list-style-type: none"> <li>Gain = <math>\frac{V_{\text{coil}}}{V_A} = \frac{-10^5}{R}</math> (Identifies gain relationship but fails to equate correctly)</li> </ul> OR <ul style="list-style-type: none"> <li>Fails to substitute correctly</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Gain = <math>\frac{V_{\text{coil}}}{V_A}</math></li> </ul> OR <ul style="list-style-type: none"> <li><math>\frac{-R}{10^5}</math> OR in words</li> </ul>                                                                     | 1     |

**Question 32 (e)**

*Outcomes assessed: H4, H7*

**MARKING GUIDELINES**

| Criteria                                                                                                                                                                                                                                                                                                | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Well constructed and extensive answer which both describes and compares the physical principles and relates them to the operation of the transducers</li> </ul>                                                                                                  | 7–8   |
| <ul style="list-style-type: none"> <li>Describes the physical principles underlying the operation of both transducers but does not compare these</li> </ul> OR <ul style="list-style-type: none"> <li>Describes one and has a comparison</li> </ul>                                                     | 5–6   |
| <ul style="list-style-type: none"> <li>Correct identification of both input and output transducers</li> </ul> AND <ul style="list-style-type: none"> <li>Describes the physical principles of one transducers</li> </ul> OR <ul style="list-style-type: none"> <li>A statement of comparison</li> </ul> | 3–4   |
| <ul style="list-style-type: none"> <li>Correctly identifies the input transducer (solar cell) and/or output transducer (ammeter)</li> </ul>                                                                                                                                                             | 1–2   |